

Volatility Modeling and Risk Forecasting with GARCH

1. Introduction & Summary

This project analyzes volatility in the S&P 500, Euro Stoxx 50, Nikkei 225, and FTSE 100. It uses daily closing prices from Yahoo Finance to compute log returns, examine volatility clustering through EDA, and perform preliminary analysis that motivates GARCH modeling. The project then estimates AR(1)-GARCH(1,1) models with Student-t innovations to model conditional volatility, generate volatility forecasts, and evaluate 99% Value-at-Risk performance through an out-of-sample backtest.

The results show that volatility is highly persistent across all four indices. The S&P 500 has the highest estimated persistence, with $\alpha[1] + \beta[1] = 0.9993$, suggesting near-integrated volatility dynamics. This may be attributed to the highly developed nature of U.S. financial markets, where a wide range of derivatives can contribute to leverage effects and contagion, leading to more persistent shocks.

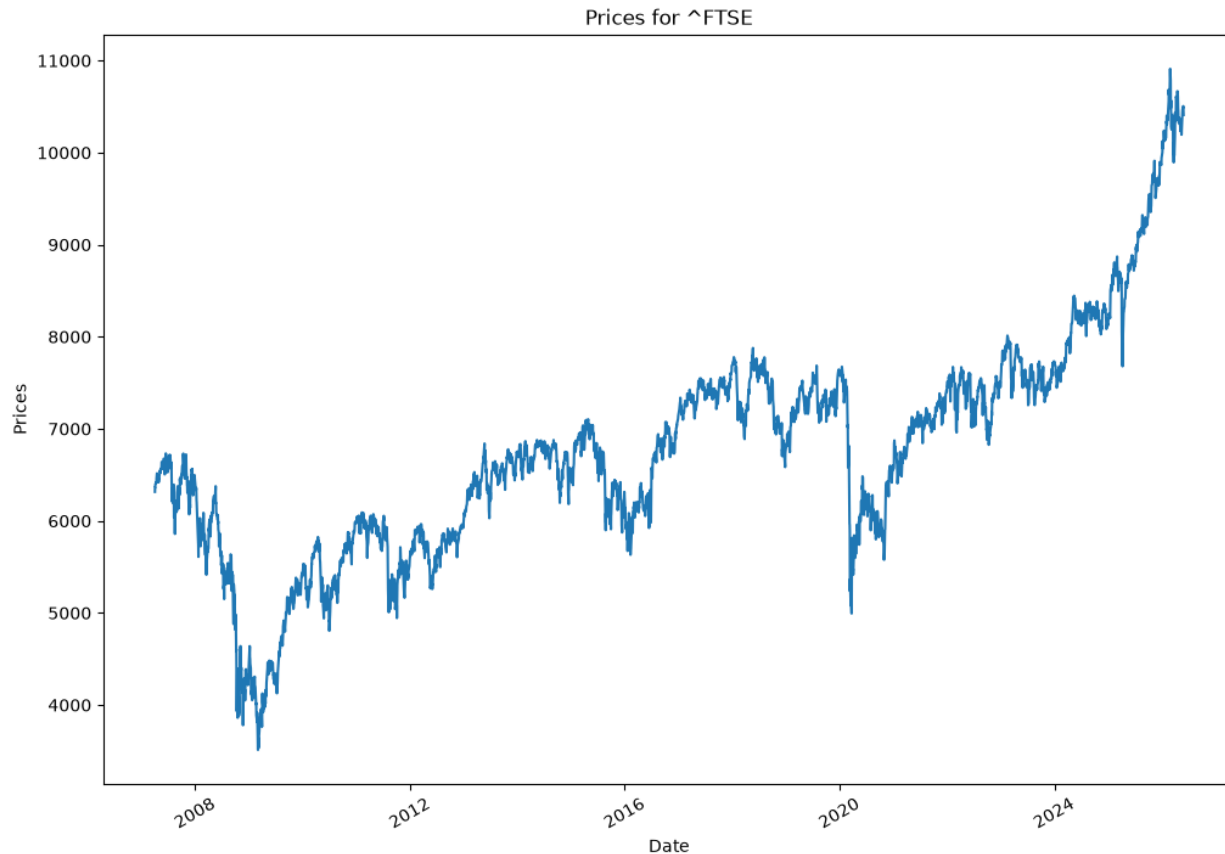
As a robustness check, I also compared the baseline AR(1)-GARCH(1,1)-t model with an AR(1)-GARCH(2,2)-t model. Although GARCH(2,2) slightly improves the AIC for the S&P 500, the BIC favors the simpler GARCH(1,1) specification for all four indices, and the residual diagnostics do not materially improve. Therefore, the AR(1)-GARCH(1,1)-t model is used as the preferred model.

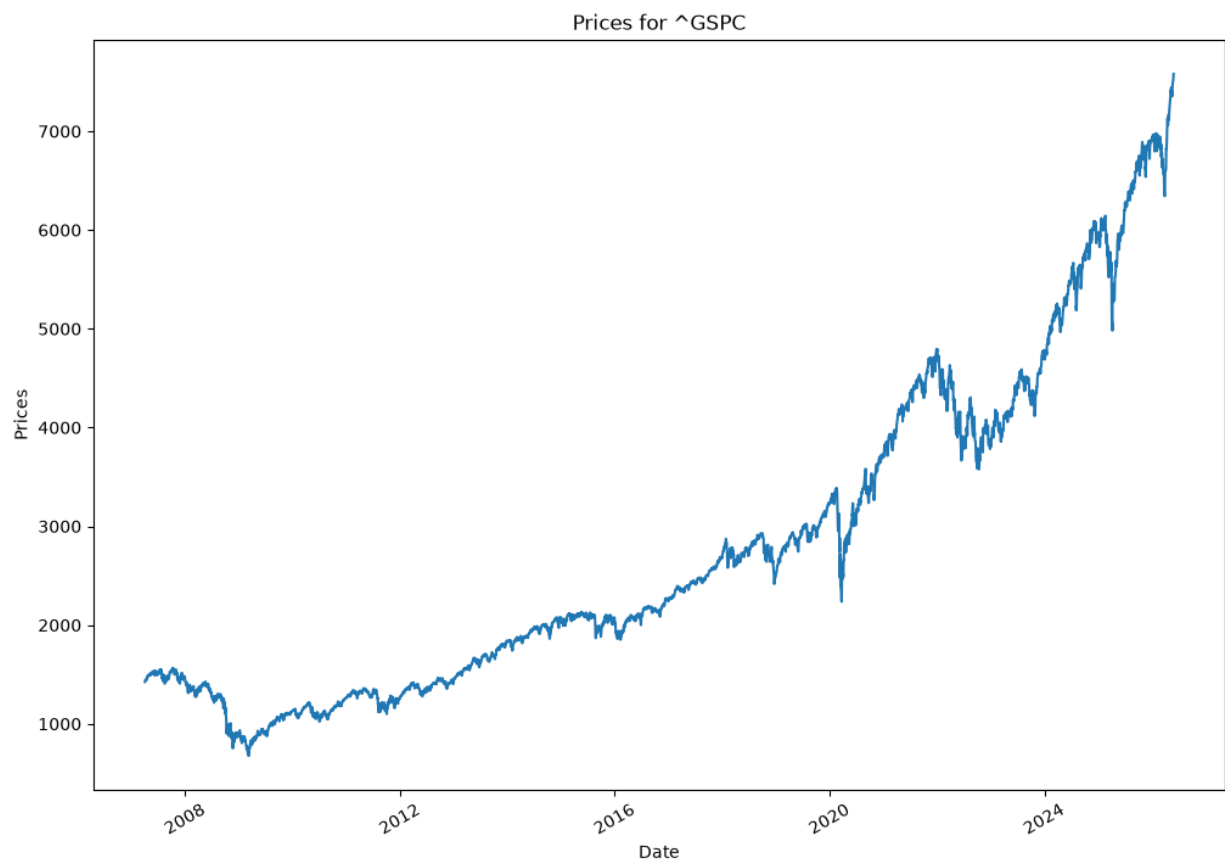
2. Data

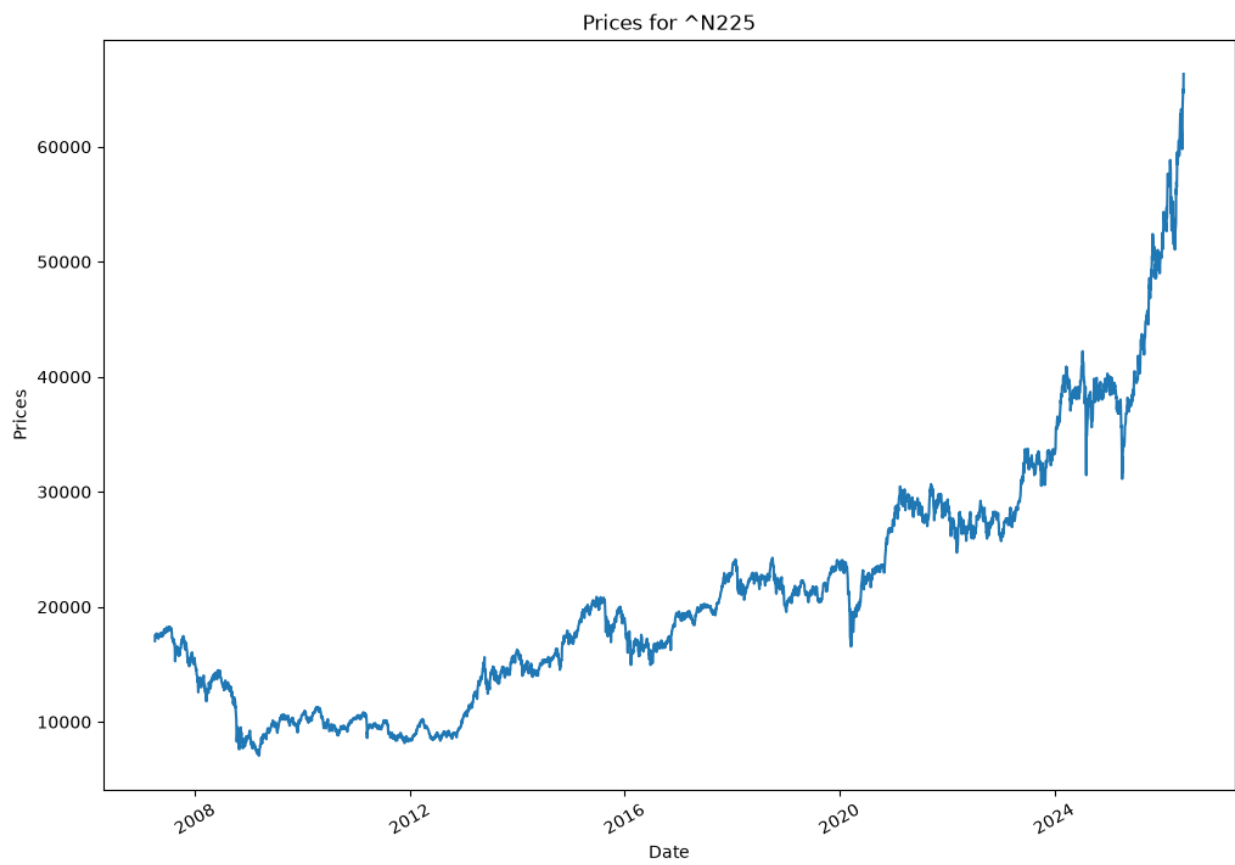
The data are obtained from Yahoo Finance using the yfinance package and include daily closing prices from 2007-04-01 through 2026-06-01.

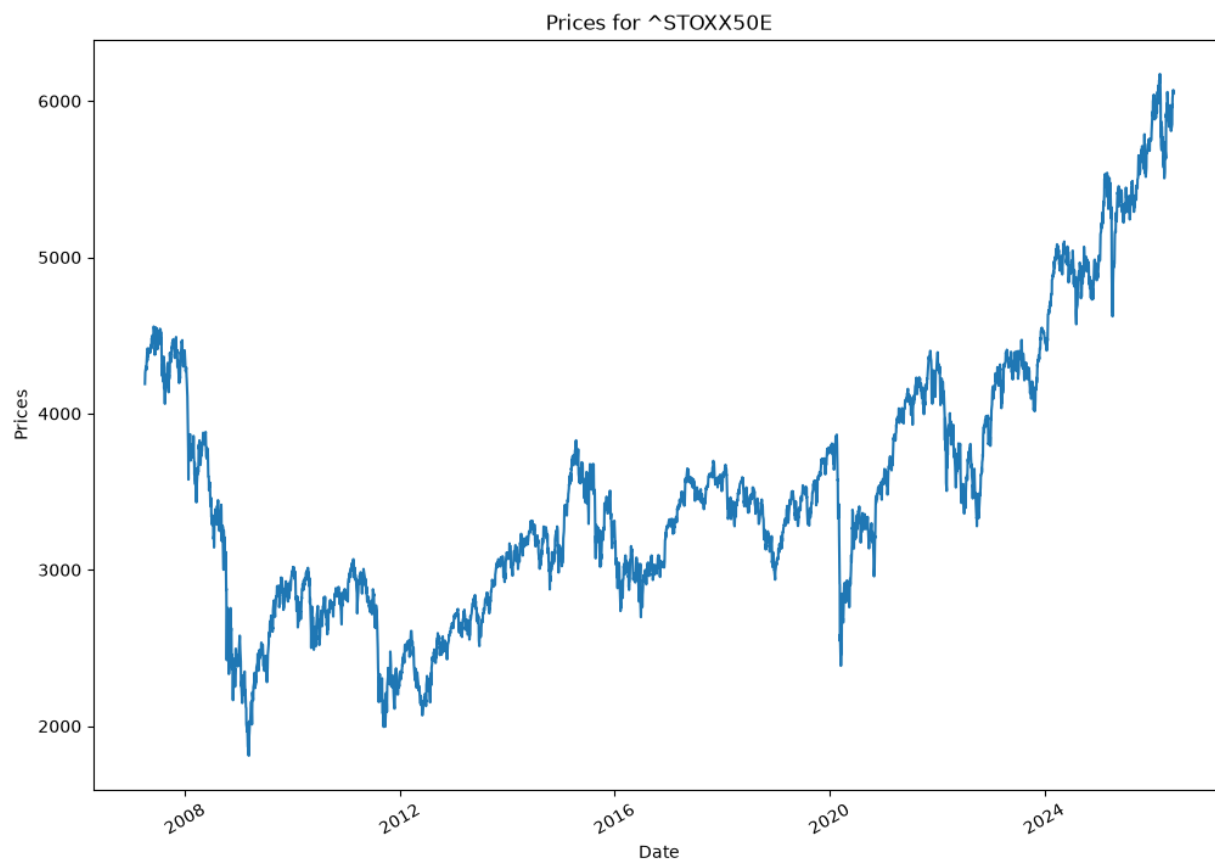
3. Exploratory Data Analysis

3.1 Price Series



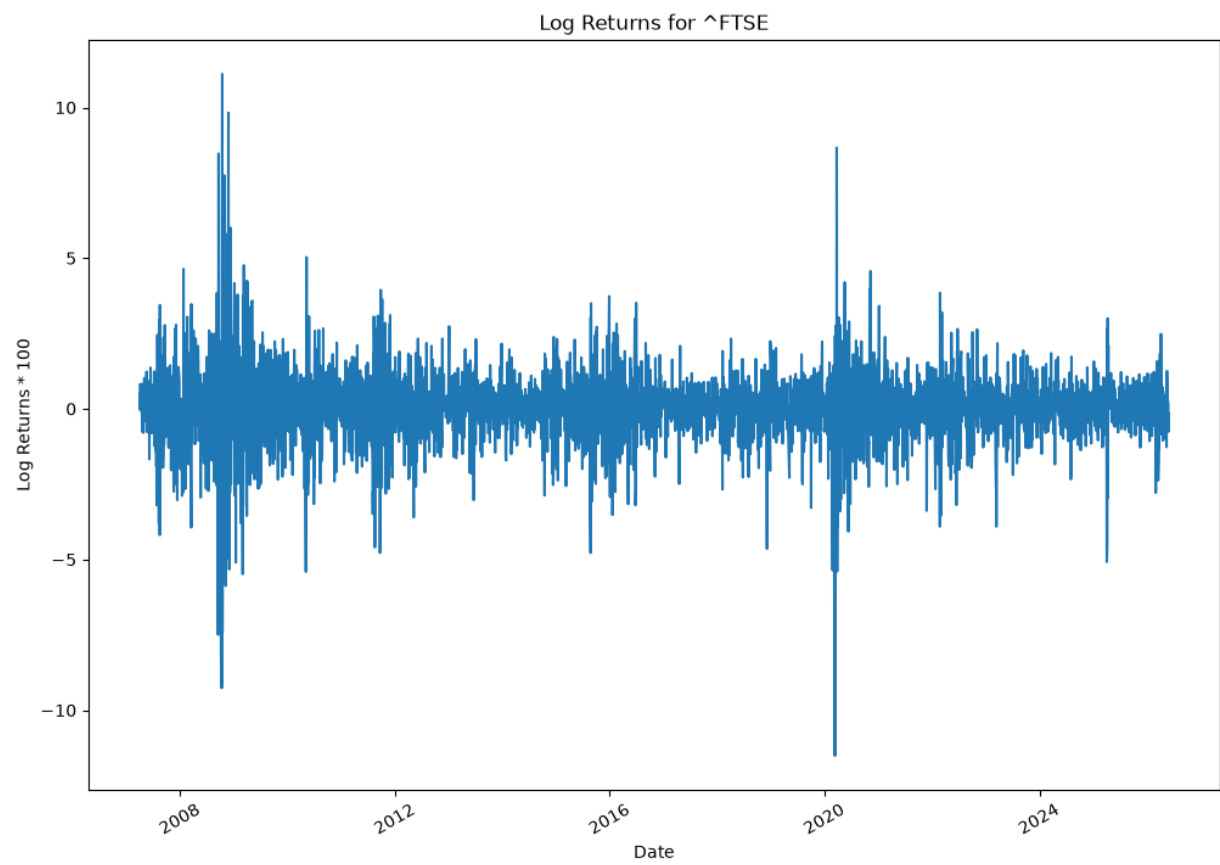


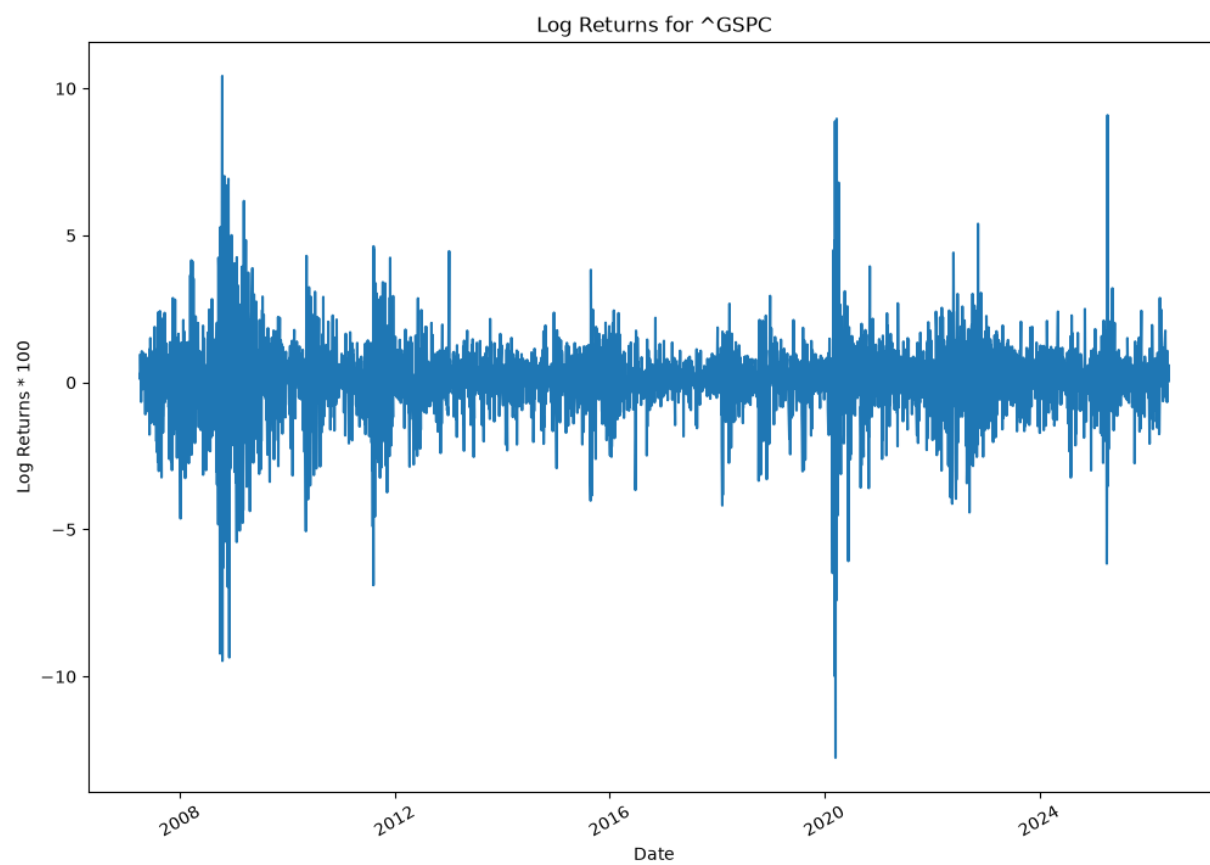


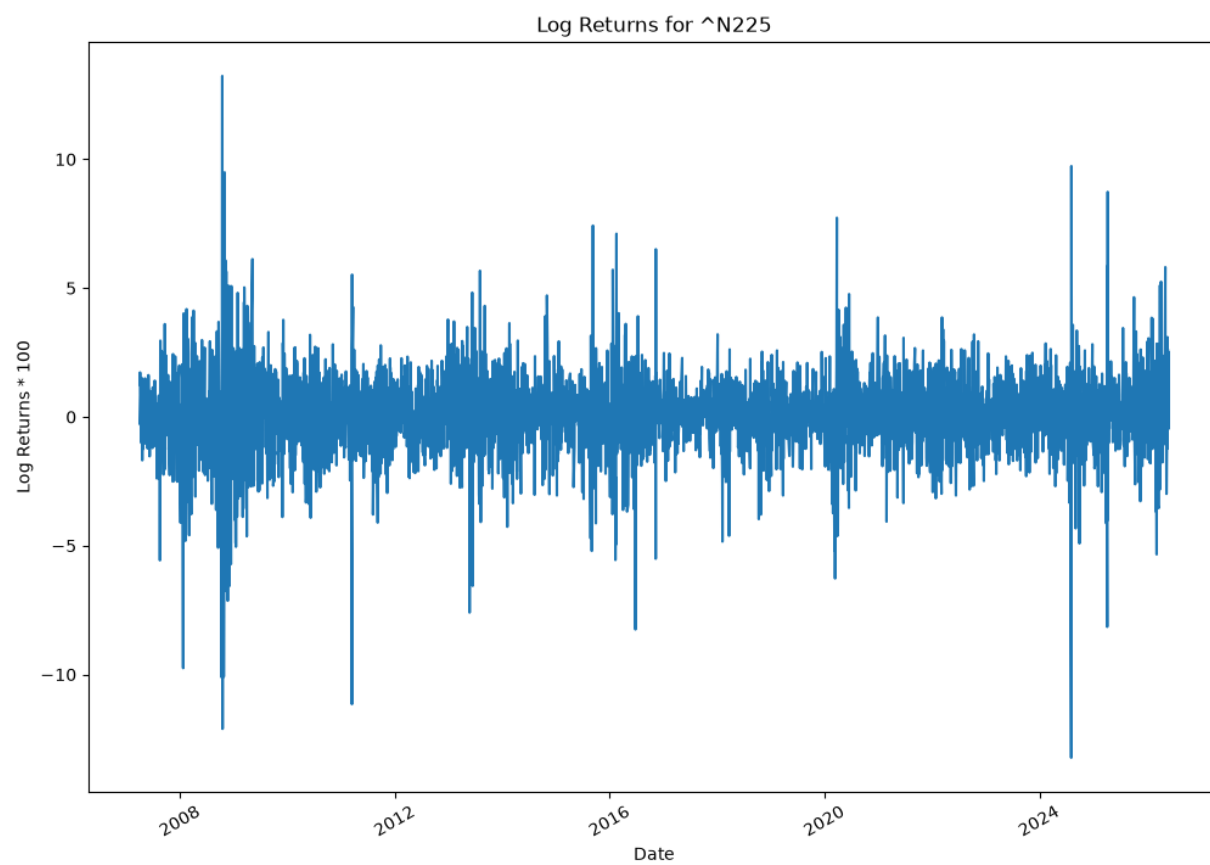


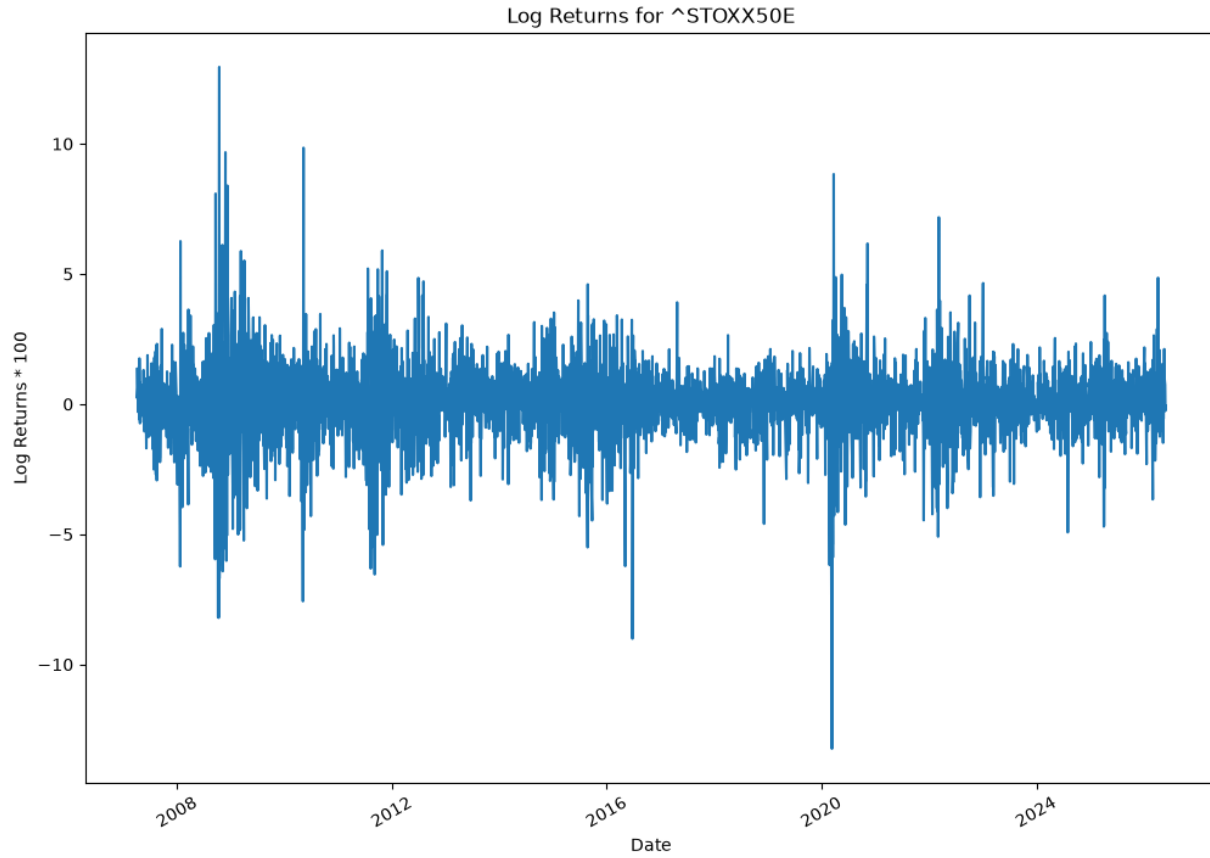
3.2 Return Series

The log return plots show that returns fluctuate around zero but exhibit clear periods of large clustered shocks (e.g. 2008 financial crisis and 2020 COVID-19 pandemic).





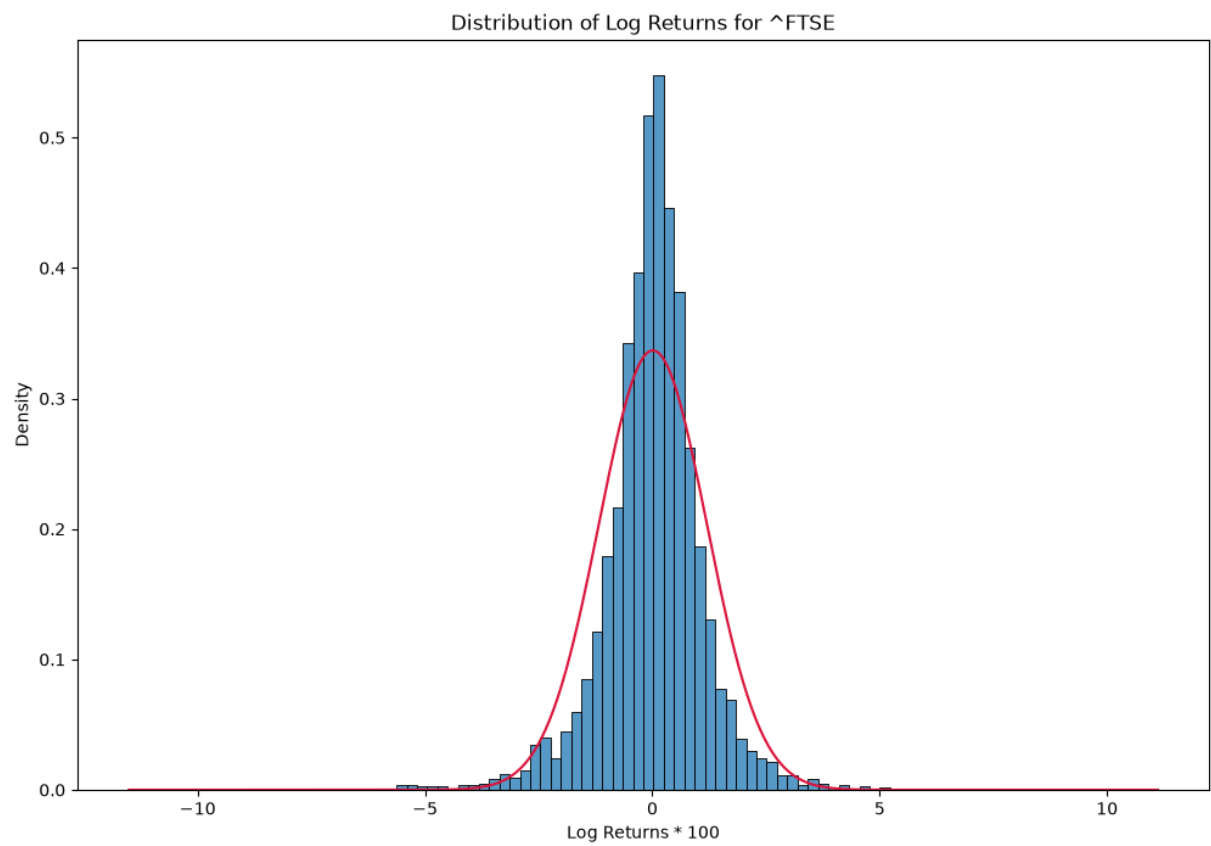


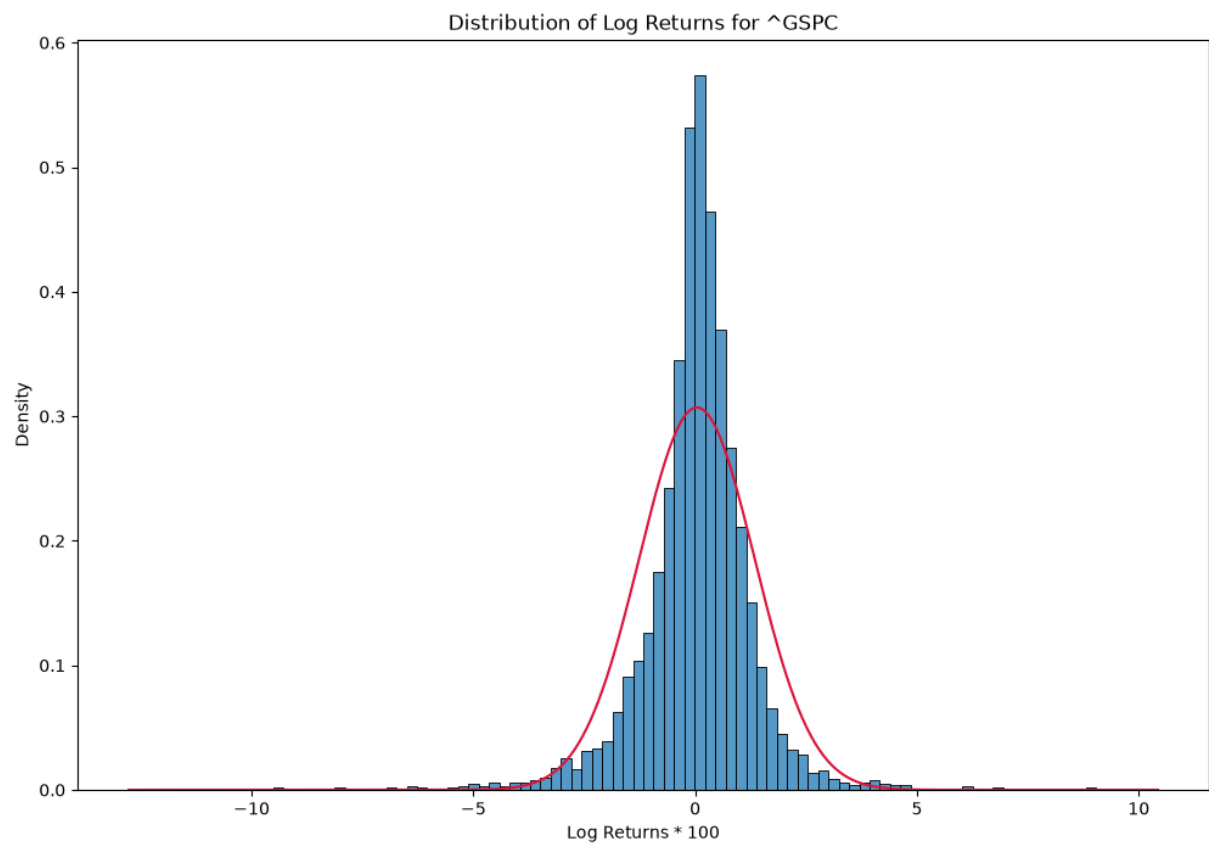


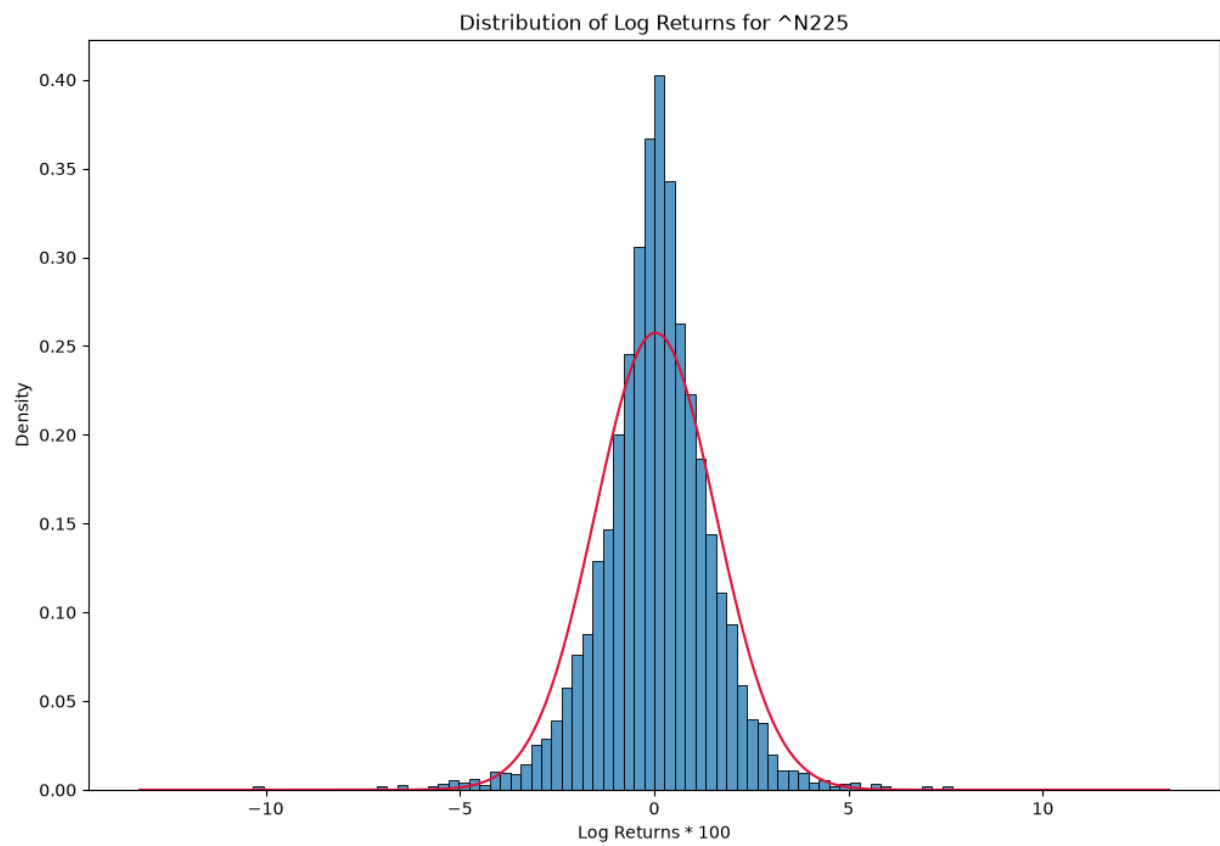
3.3 Return Distributions

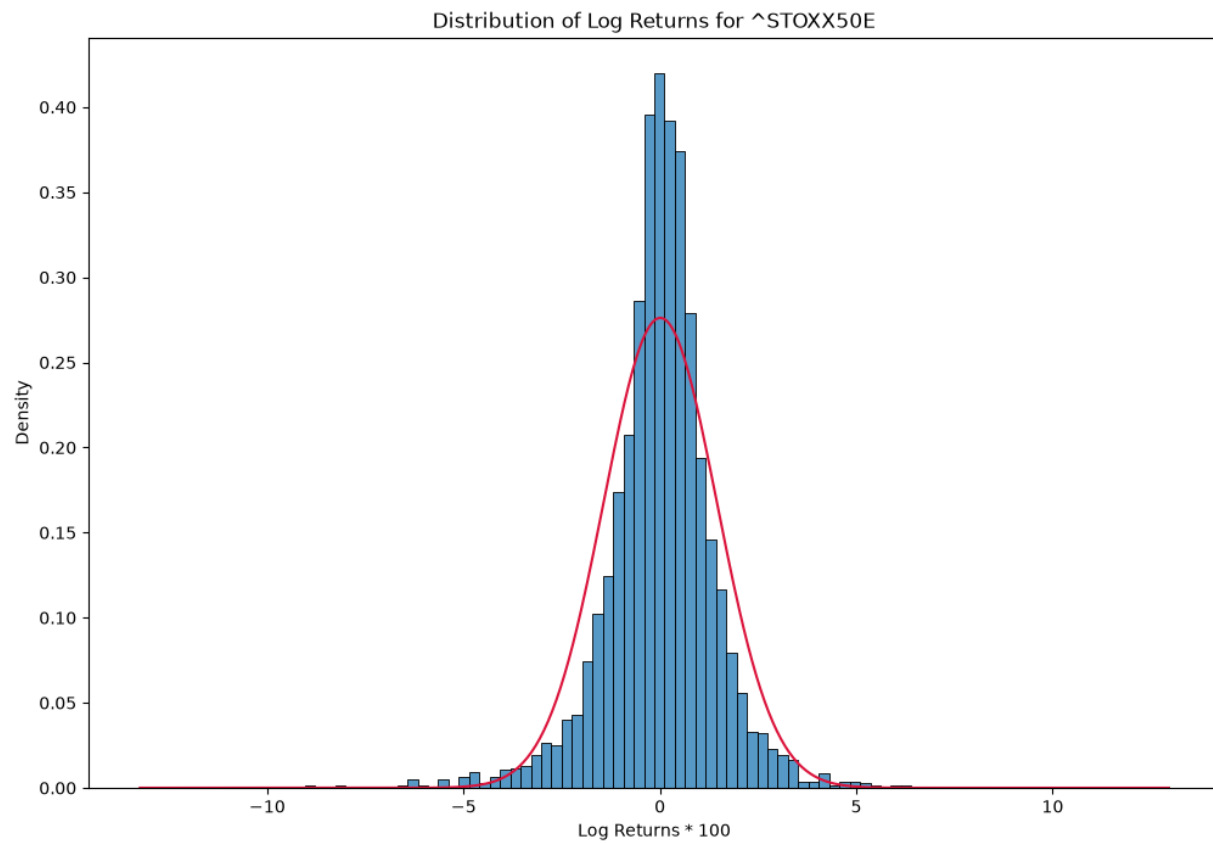
The return distributions are centered close to zero but show negative skewness and high kurtosis, indicating heavier tails than a normal distribution.

Index	Mean	Std. Dev.	Skewness	Kurtosis
FTSE	0.0114	1.1844	-0.2994	11.1410
GSPC	0.0382	1.2994	-0.6253	11.1280
N225	0.0311	1.5496	-0.4663	8.3043
STOXX50E	0.0084	1.4447	-0.2565	8.0194



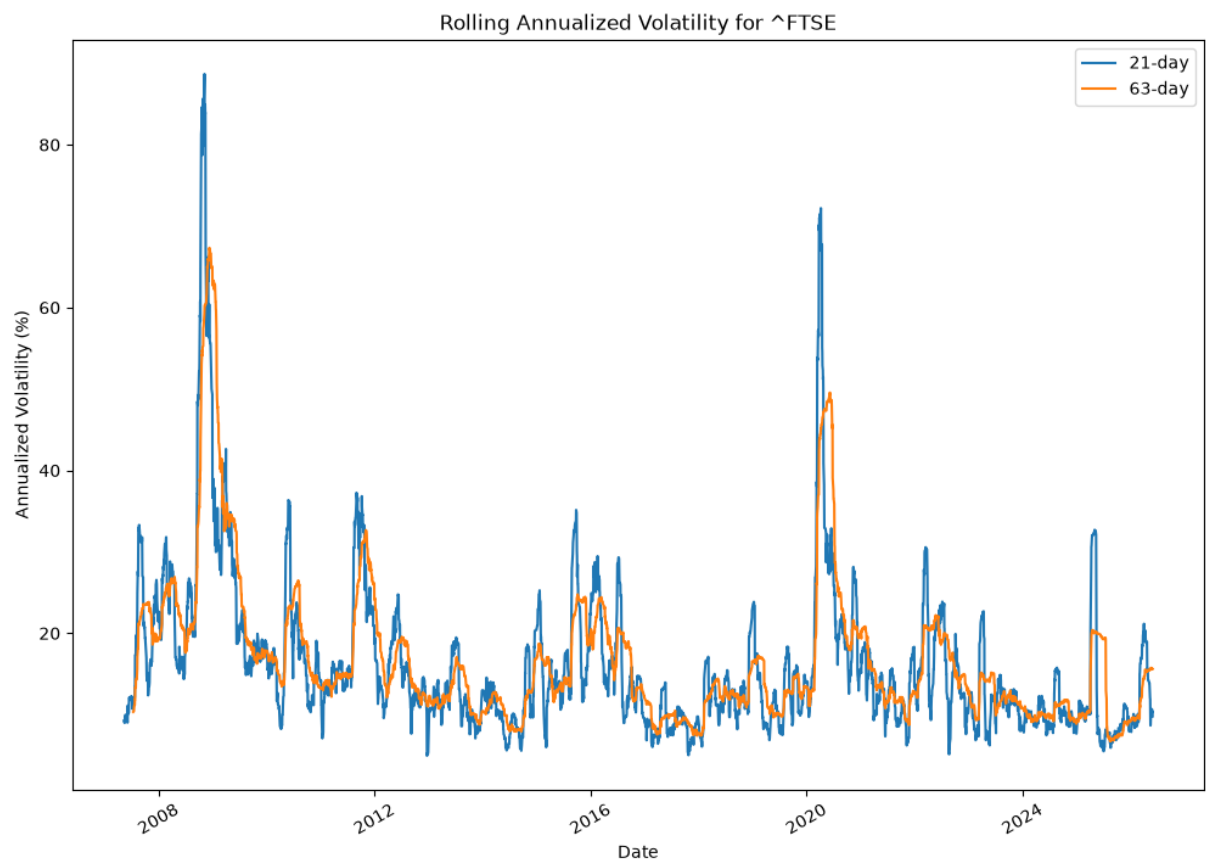


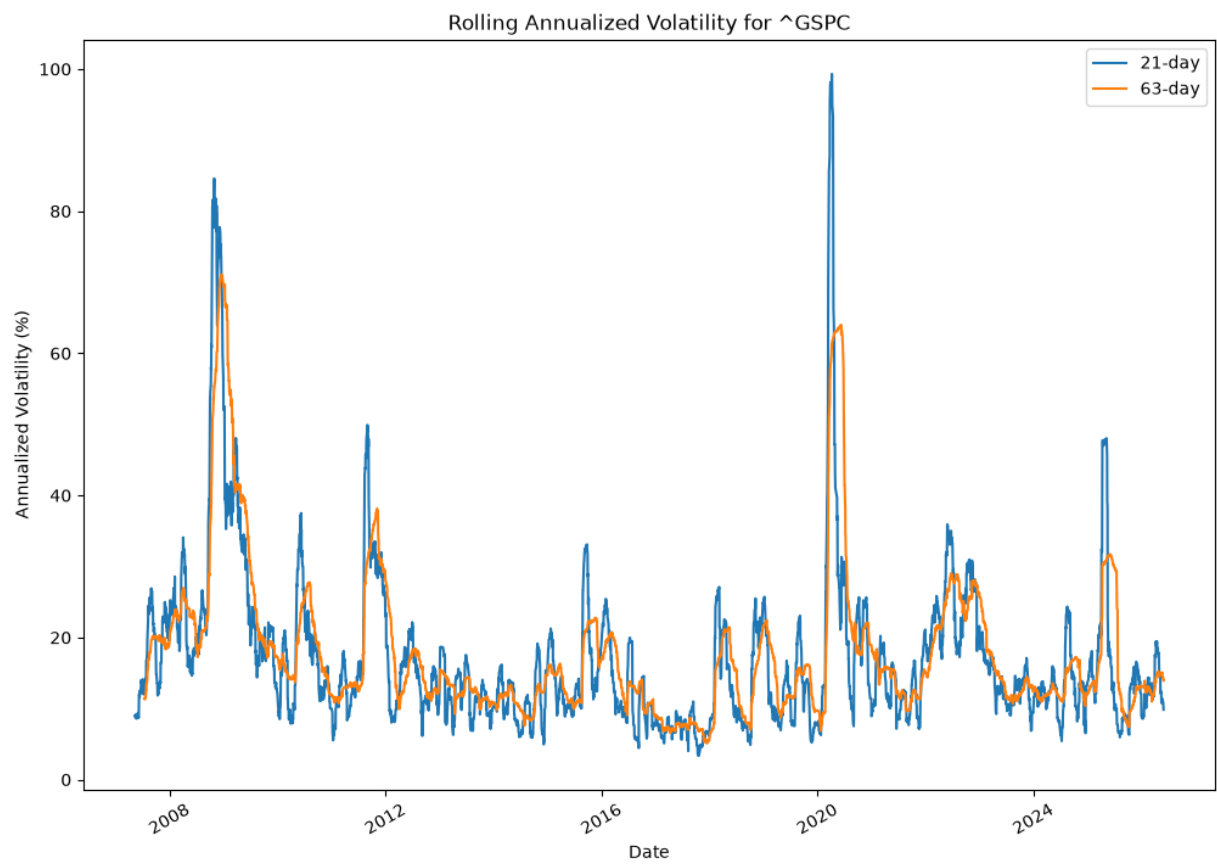


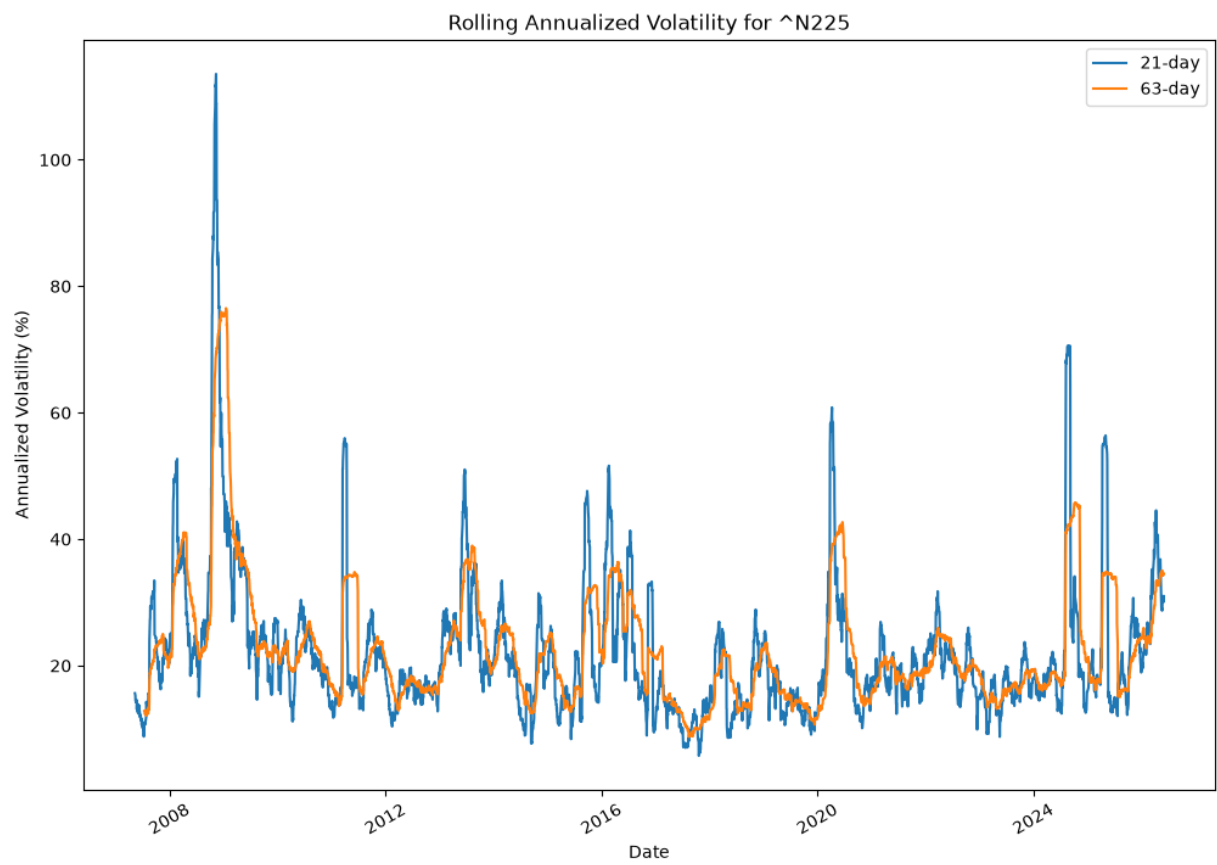


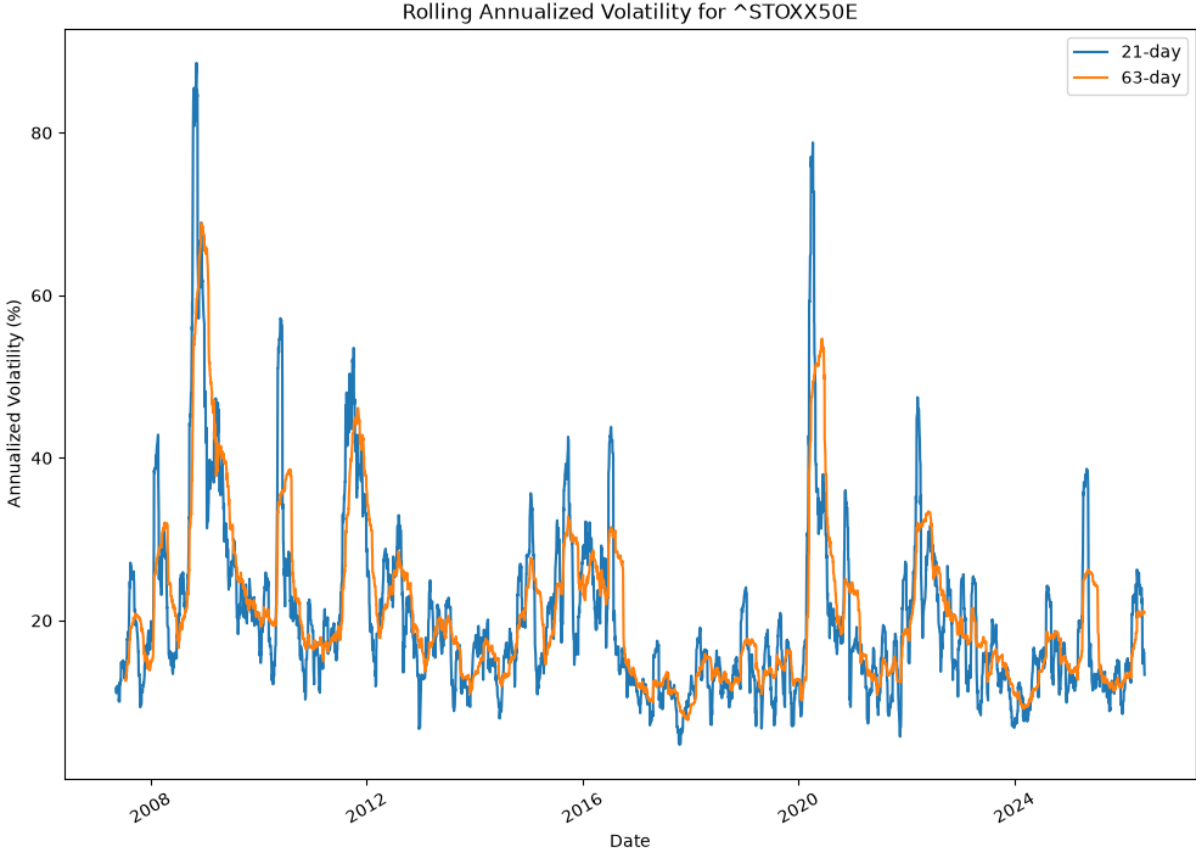
3.4 Rolling Volatility

The rolling volatility plots once again show that volatility changes over time and tends to remain elevated after large market shocks.









4. Preliminary Tests

4.1 Ljung-Box Tests on Log Returns

The p-value of the Ljung-Box Tests on squared log returns is very small (numerically zero), which indicates volatility clustering and suggests GARCH modeling.

index	series	lag	lb_stat	lb_pvalue
^FTSE	returns	10	47.7769	6.817e-07
^FTSE	returns	20	63.5542	1.979e-06
^FTSE	squared_returns	10	2813	0
^FTSE	squared_returns	20	4199	0
^GSPC	returns	10	137.7233	1.233e-24
^GSPC	returns	20	184.5889	1.224e-28
^GSPC	squared_returns	10	5061	0
^GSPC	squared_returns	20	6666	0
^N225	returns	10	25.1781	0.005
^N225	returns	20	51.4986	0.0001347
^N225	squared_returns	10	2012	0
^N225	squared_returns	20	2642	0
^STOXX50E	returns	10	34.4138	0.0001571
^STOXX50E	returns	20	45.2548	0.001
^STOXX50E	squared_returns	10	1690	0
^STOXX50E	squared_returns	20	2409	0

4.2 Stationarity Tests

The ADF tests reject the unit-root null for time series of all four indices, so the stationarity condition is satisfied for GARCH modeling.

index	ADF	p-value	lags	obs	1% crit	5% crit	10% crit
^FTSE	-28.0801	0	5	4370	-3.4318	-2.8622	-2.5671
^GSPC	-15.6464	1.617e-28	17	4358	-3.4319	-2.8622	-2.5671
^N225	-70.3178	0	0	4375	-3.4318	-2.8622	-2.5671
^STOXX50E	-31.4142	0	4	4371	-3.4318	-2.8622	-2.5671

4.3 ARCH LM Tests

The ARCH LM tests show very strong evidence of ARCH effects in all four return series after fitting an AR(1) mean model. This suggests that a GARCH-type model is appropriate for modeling these series.

index	lags	LM stat	LM p	F stat	F p
^FTSE	10	961.7744	3.198e-200	123.0391	1.086e-226
^GSPC	10	1235	4.179e-259	171.7683	2.33e-305
^N225	10	785.6011	2.569e-162	95.5562	2.518e-179
^STOXX50E	10	675.4491	1.168e-138	79.7057	6.234e-151

5. Model Specification

The fitted model is specified as:

- Mean equation: $r_t = \mu + \phi r_{t-1} + \varepsilon_t$
- Shock decomposition: $\varepsilon_t = \sigma_t z_t$
- Variance equation: $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
- Innovation distribution: $z_t \sim t_\nu$

6. Estimation Results

6.1 Model Fit

index	log_likelihood	AIC	BIC
^FTSE	-5909	1.183e+04	1.187e+04
^GSPC	-6017	1.205e+04	1.208e+04
^N225	-7392	1.48e+04	1.483e+04
^STOXX50E	-6939	1.389e+04	1.393e+04

6.2 Parameter Estimates

For all four indices, the AR(1)-GARCH(1, 1) model shows highly significant volatility clustering with persistent GARCH effects and fat-tailed return shocks.

index	parameter	estimate	std_error	t_stat	p_value
^FTSE	Const	0.0521	0.0118	4.4105	1.031e-05
^FTSE	^FTSE[1]	-0.0056	0.0154	-0.3646	0.7154
^FTSE	omega	0.0311	0.0071	4.3921	1.123e-05
^FTSE	alpha[1]	0.1311	0.0192	6.8379	8.037e-12
^FTSE	beta[1]	0.8485	0.021	40.459	0
^FTSE	nu	5.8568	0.5162	11.3469	7.686e-30
^GSPC	Const	0.1104	0.0113	9.7755	1.435e-22
^GSPC	^GSPC[1]	-0.0575	0.0147	-3.9107	9.202e-05
^GSPC	omega	0.021	0.0047	4.5121	6.42e-06
^GSPC	alpha[1]	0.1471	0.0151	9.7149	2.604e-22
^GSPC	beta[1]	0.8522	0.0132	64.4541	0
^GSPC	nu	4.9918	0.3798	13.1448	1.823e-39
^N225	Const	0.0834	0.0174	4.8091	1.516e-06
^N225	^N225[1]	-0.0392	0.0155	-2.5382	0.0111
^N225	omega	0.0645	0.0169	3.819	0.000134

index	parameter	estimate	std_error	t_stat	p_value
$\hat{N}225$	alpha[1]	0.1144	0.0184	6.2288	4.7e-10
$\hat{N}225$	beta[1]	0.8597	0.022	39.0374	0
$\hat{N}225$	nu	6.5603	0.6572	9.9818	1.831e-23
$\hat{STOXX50E}$	Const	0.0762	0.0153	4.9783	6.416e-07
$\hat{STOXX50E}$	$\hat{STOXX50E}[1]$	-0.0331	0.0149	-2.2292	0.0258
$\hat{STOXX50E}$	omega	0.0353	0.0103	3.4067	0.0006576
$\hat{STOXX50E}$	alpha[1]	0.1098	0.0192	5.7258	1.03e-08
$\hat{STOXX50E}$	beta[1]	0.8781	0.0206	42.5703	0
$\hat{STOXX50E}$	nu	5.3101	0.4259	12.4685	1.109e-35

6.3 Volatility Persistence

The large values of $\alpha[1] + \beta[1]$ suggest high volatility persistence across four indices, with $\hat{GSPC}$ showing near-integrated volatility persistence at 0.9993, followed by $\hat{STOXX50E}$, $\hat{FTSE}$, and $\hat{N}225$.

index	omega	alpha[1]	beta[1]	alpha_beta
$\hat{GSPC}$	0.021	0.1471	0.8522	0.9993
$\hat{STOXX50E}$	0.0353	0.1098	0.8781	0.988
$\hat{FTSE}$	0.0311	0.1311	0.8485	0.9795
$\hat{N}225$	0.0645	0.1144	0.8597	0.974

6.4 Model Comparison

As a robustness check, I compare the baseline AR(1)-GARCH(1,1)-t model with an AR(1)-GARCH(2,2)-t model. The higher-order model improves the AIC only for the S&P 500, while the BIC is higher for all four indices. Additionally, the p-values of the Ljung-Box tests do not materially improve for the AR(1)-GARCH(2,2) model. Thus, the increased model complexity is not justified, and I proceed with the AR(1)-GARCH(1,1) model.

Fit Criteria

index	AIC (1,1)	BIC (1,1)	AIC (2,2)	BIC (2,2)	dAIC	dBIC
$\hat{FTSE}$	11830.08	11868.38	11834.03	11885.10	3.95	16.72
$\hat{GSPC}$	12045.97	12084.27	12040.95	12092.02	-5.02	7.75
$\hat{N}225$	14796.10	14834.40	14799.74	14850.81	3.64	16.41
$\hat{STOXX50E}$	13889.11	13927.41	13892.38	13943.45	3.27	16.04

Ljung-Box Diagnostic P-Values (Lag 20)

index	z p-value (1,1)	z p-value (2,2)	z^2 p (1,1)	z^2 p (2,2)
$\hat{FTSE}$	0.2552	0.251	0.6067	0.5858
$\hat{GSPC}$	0.0499	0.04465	0.9214	0.8204
$\hat{N}225$	0.7749	0.7675	0.001237	0.002578
$\hat{STOXX50E}$	0.2864	0.2769	0.4326	0.4814

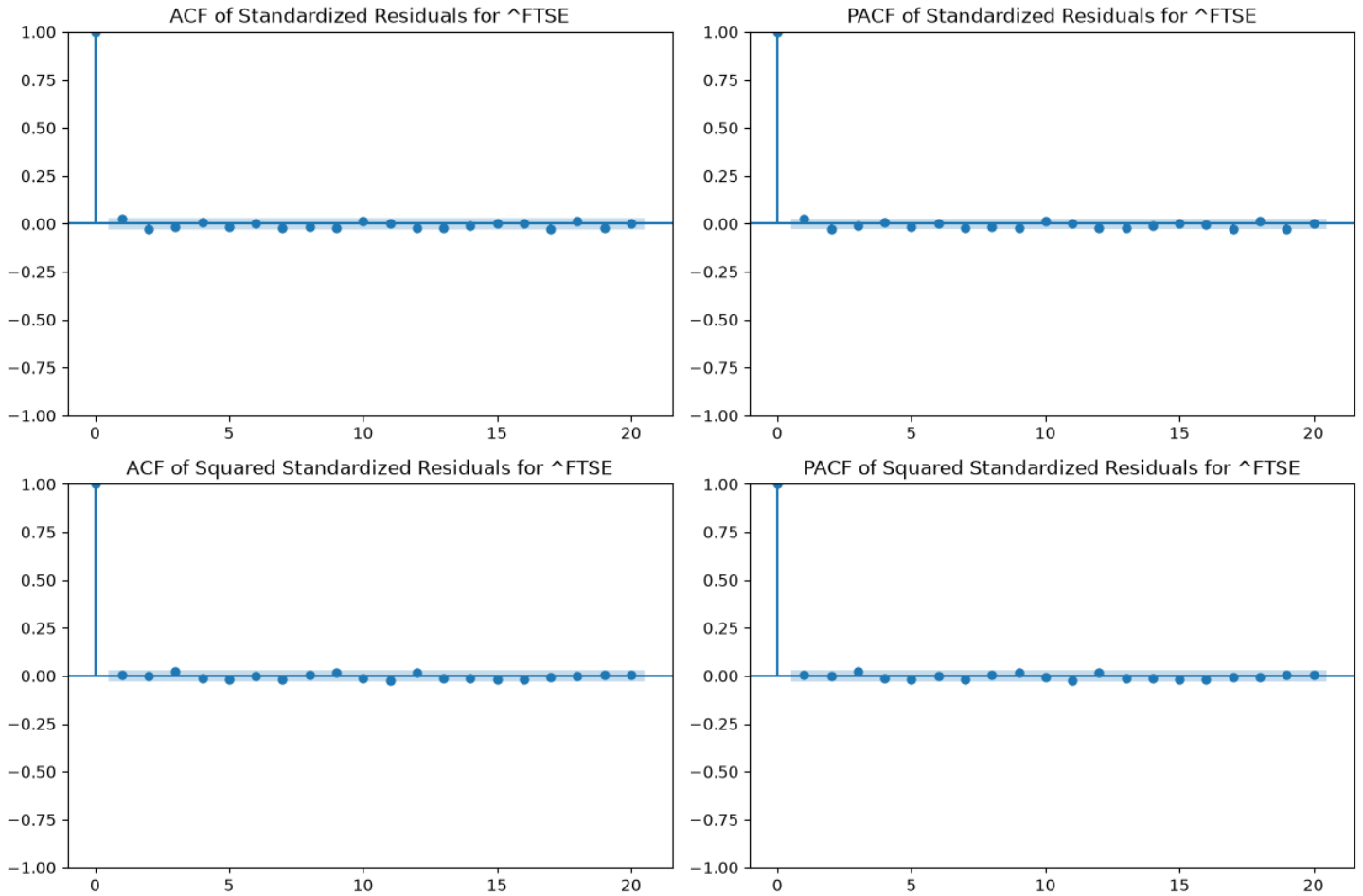
7. Model Diagnostics

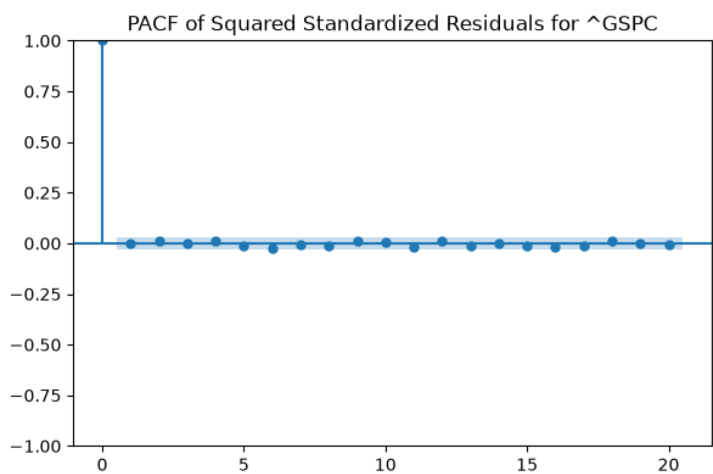
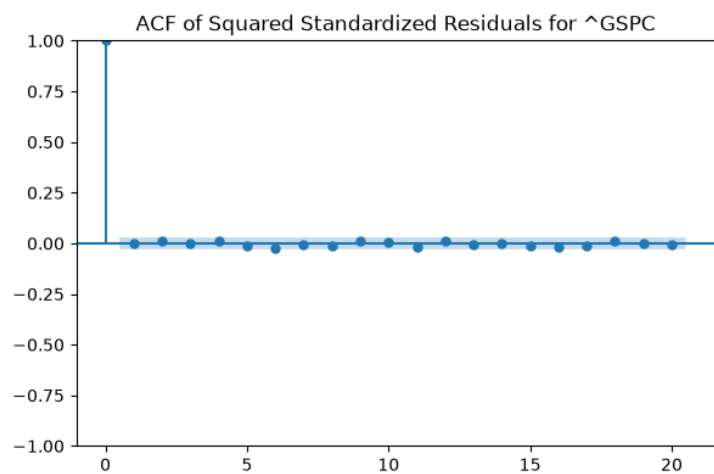
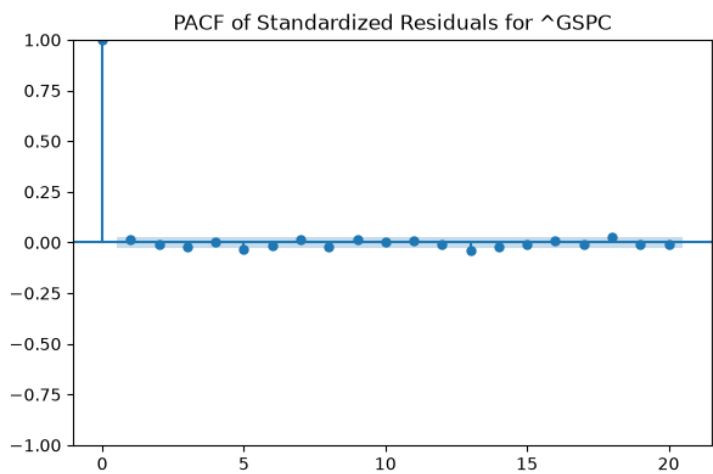
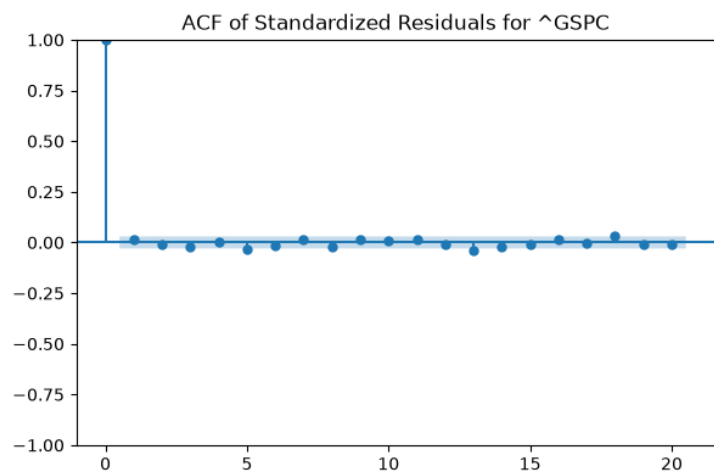
The Ljung-Box tests show that the AR(1)-GARCH(1, 1) model has sufficiently removed autocorrelation in the standardized residuals across four indices. For squared standardized residuals, the model performs well in removing autocorrelation in $\hat{FTSE}$ and $\hat{STOXX50E}$, but it does not fully capture volatility clustering for $\hat{GSPC}$ and $\hat{N}225$.

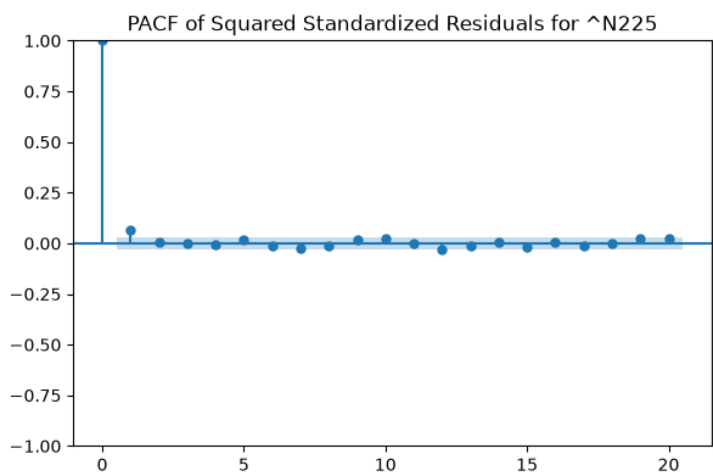
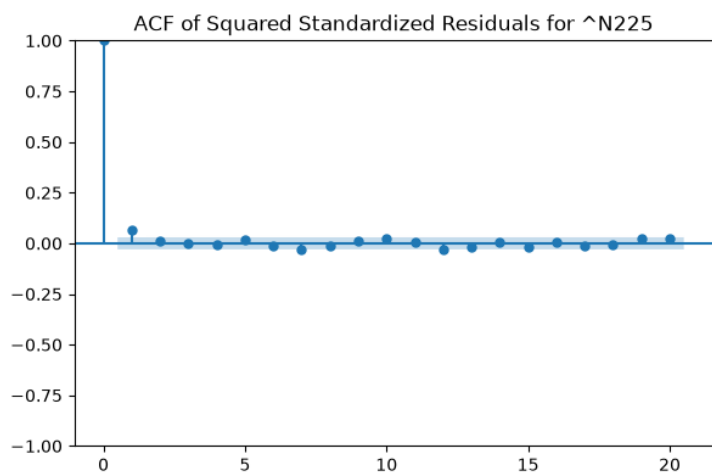
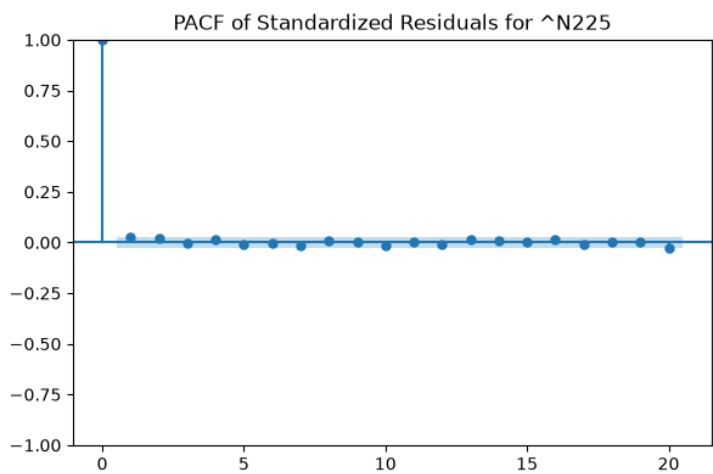
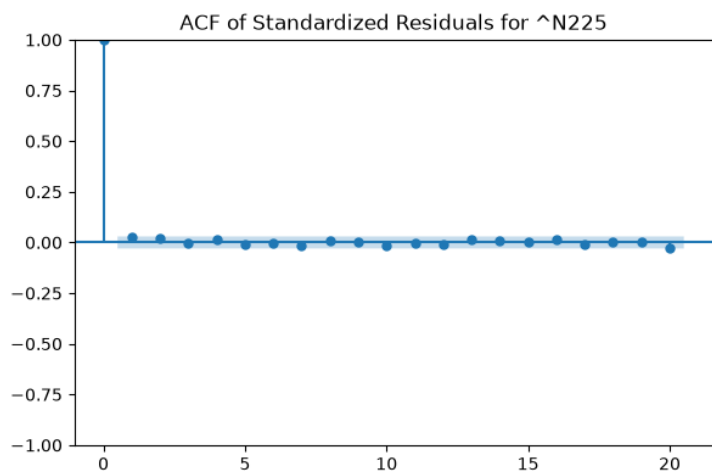
index	test	lag	stat	p_value
$\hat{FTSE}$	lb_std_resid	10	12.9257	0.2279
$\hat{FTSE}$	lb_std_resid	20	23.713	0.2552
$\hat{FTSE}$	lb_std_resid_squared	10	8.4607	0.5839

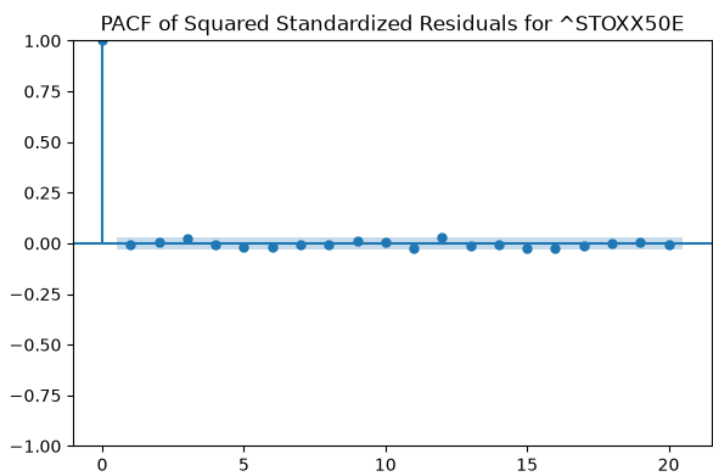
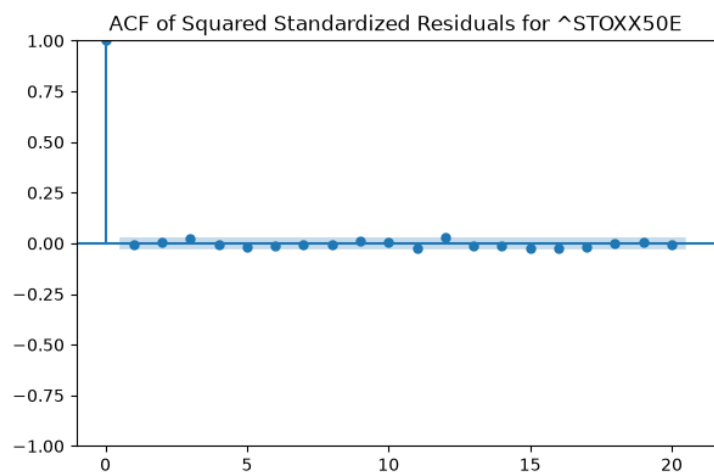
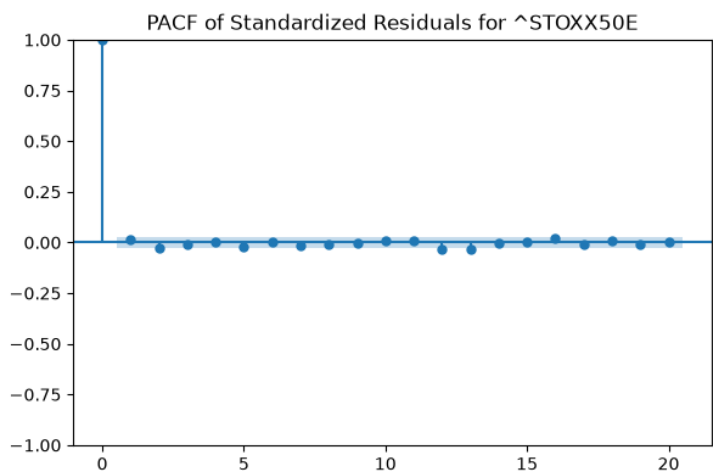
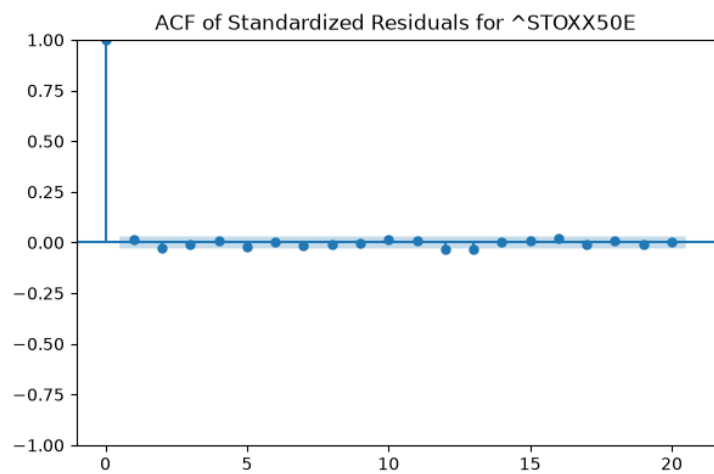
index	test	lag	stat	p_value
^FTSE	lb_std_resid_squared	20	17.7075	0.6067
^GSPC	lb_std_resid	10	14.275	0.1608
^GSPC	lb_std_resid	20	31.4188	0.0499
^GSPC	lb_std_resid_squared	10	5.6836	0.8411
^GSPC	lb_std_resid_squared	20	11.8408	0.9214
^N225	lb_std_resid	10	8.6317	0.5674
^N225	lb_std_resid	20	15.027	0.7749
^N225	lb_std_resid_squared	10	30.0278	0.0008477
^N225	lb_std_resid_squared	20	44.6328	0.0012
^STOXX50E	lb_std_resid	10	8.4025	0.5896
^STOXX50E	lb_std_resid	20	23.0496	0.2864
^STOXX50E	lb_std_resid_squared	10	5.9099	0.8228
^STOXX50E	lb_std_resid_squared	20	20.41	0.4326

The following ACF/PACF plots show the autocorrelation structure of the standardized residuals and squared standardized residuals from the fitted AR(1)-GARCH(1,1)-t models. As most autocorrelations lie within the confidence bands after lag 0, the models appear to fit relatively well, although some remaining dependence is still visible for certain squared standardized residual series.



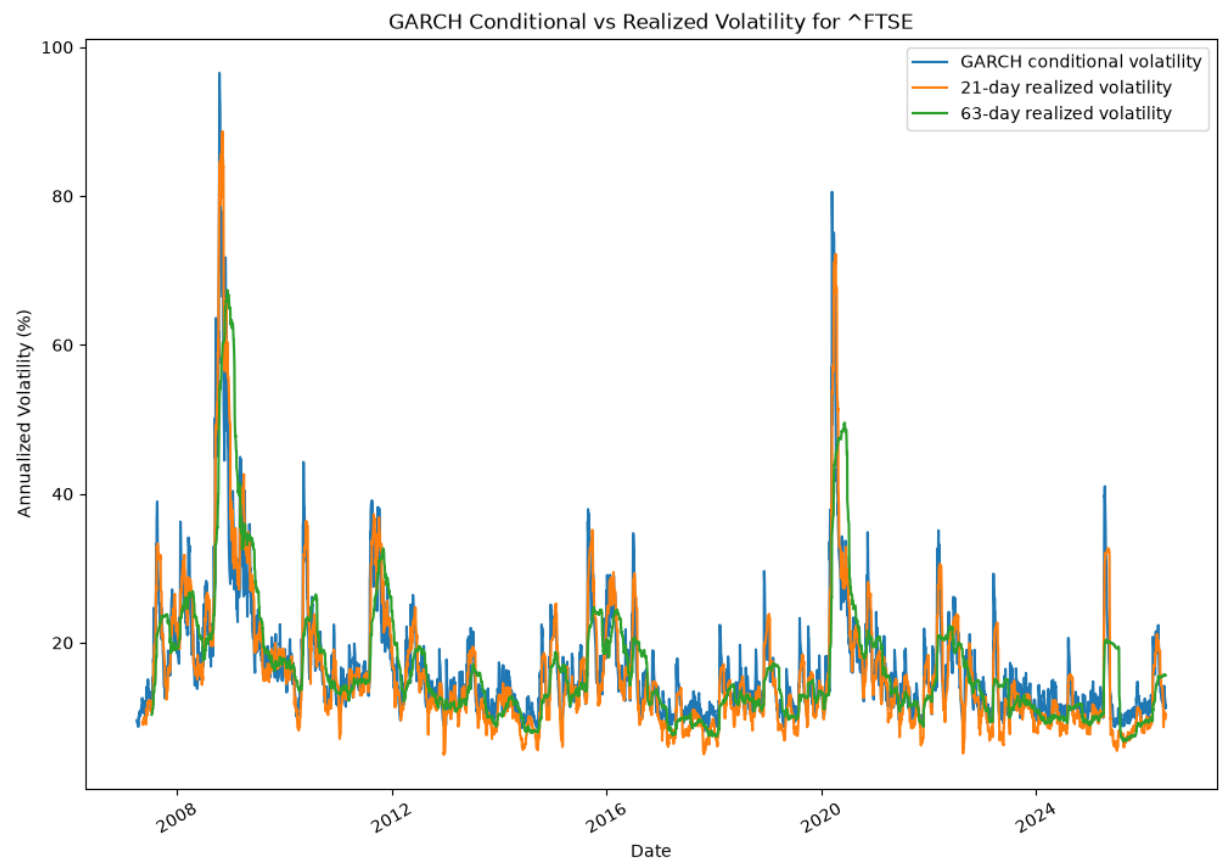


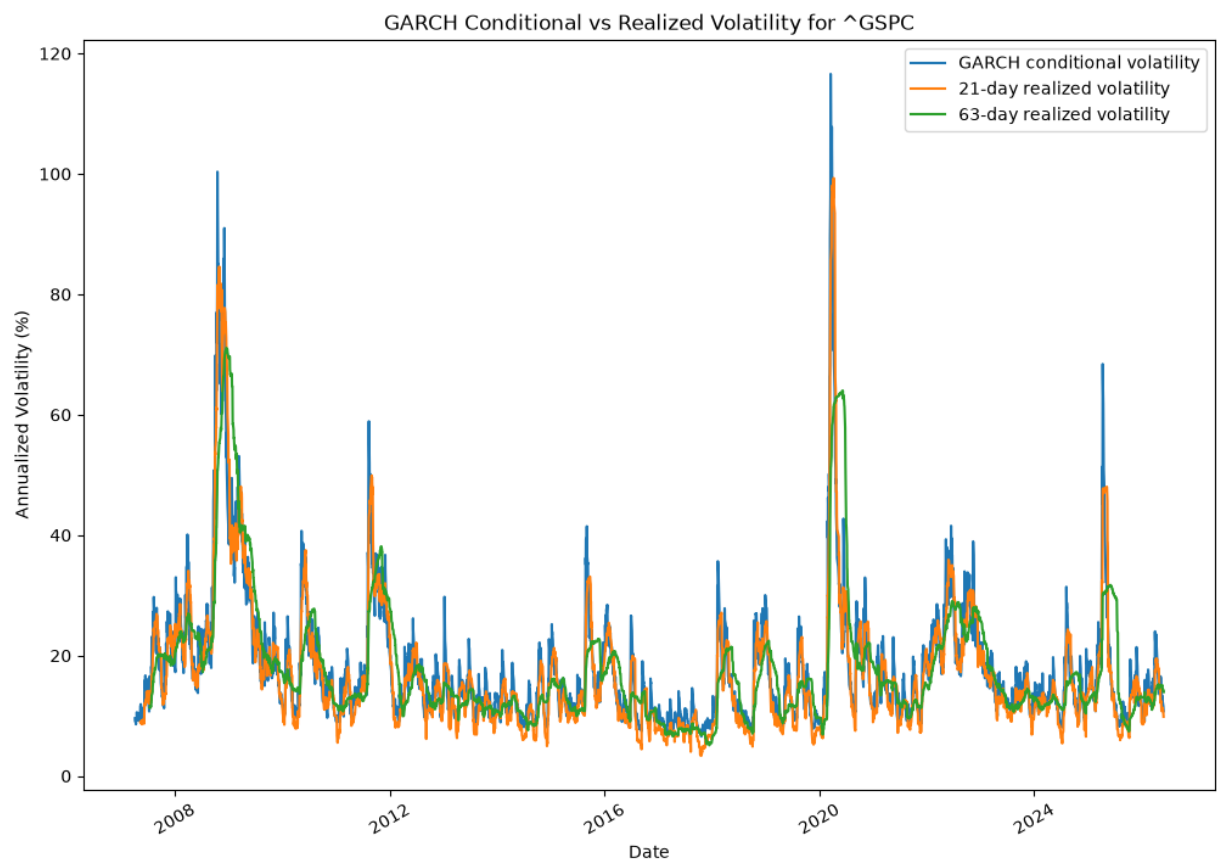


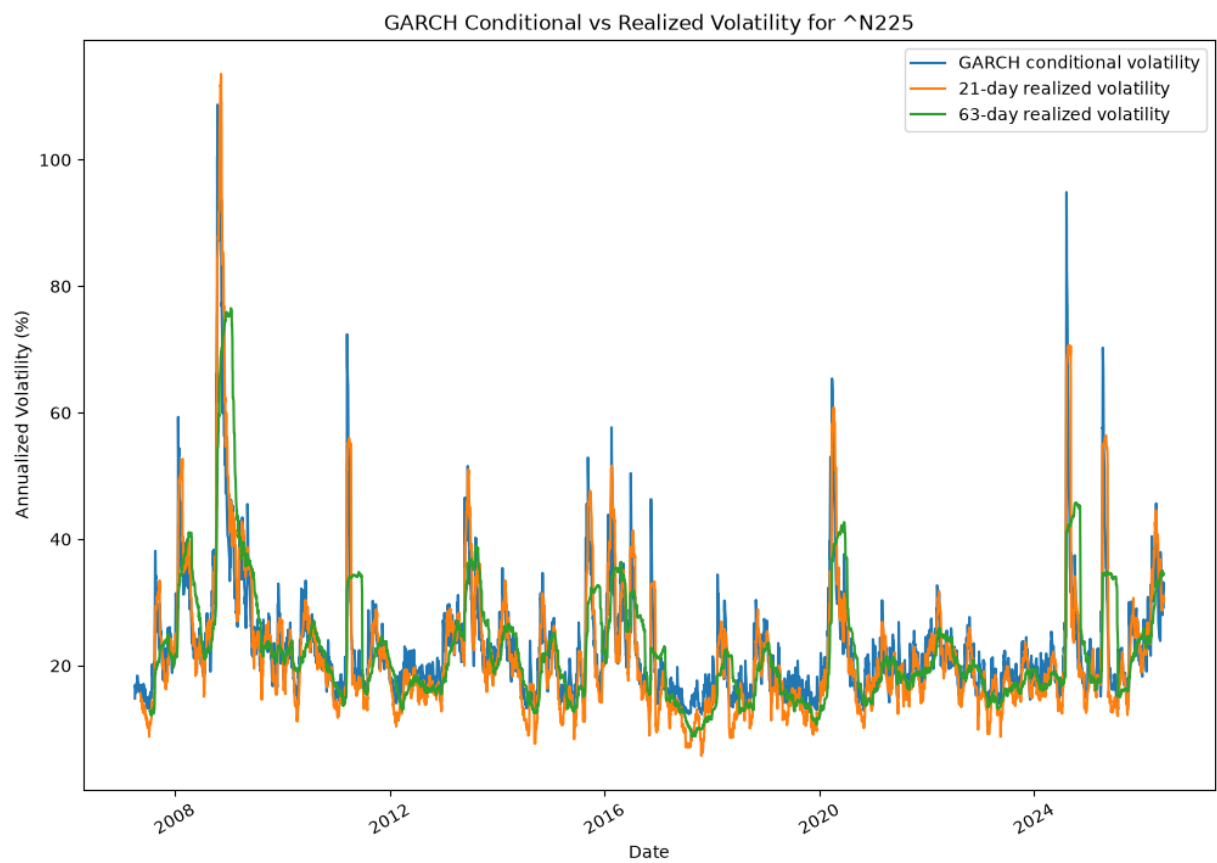


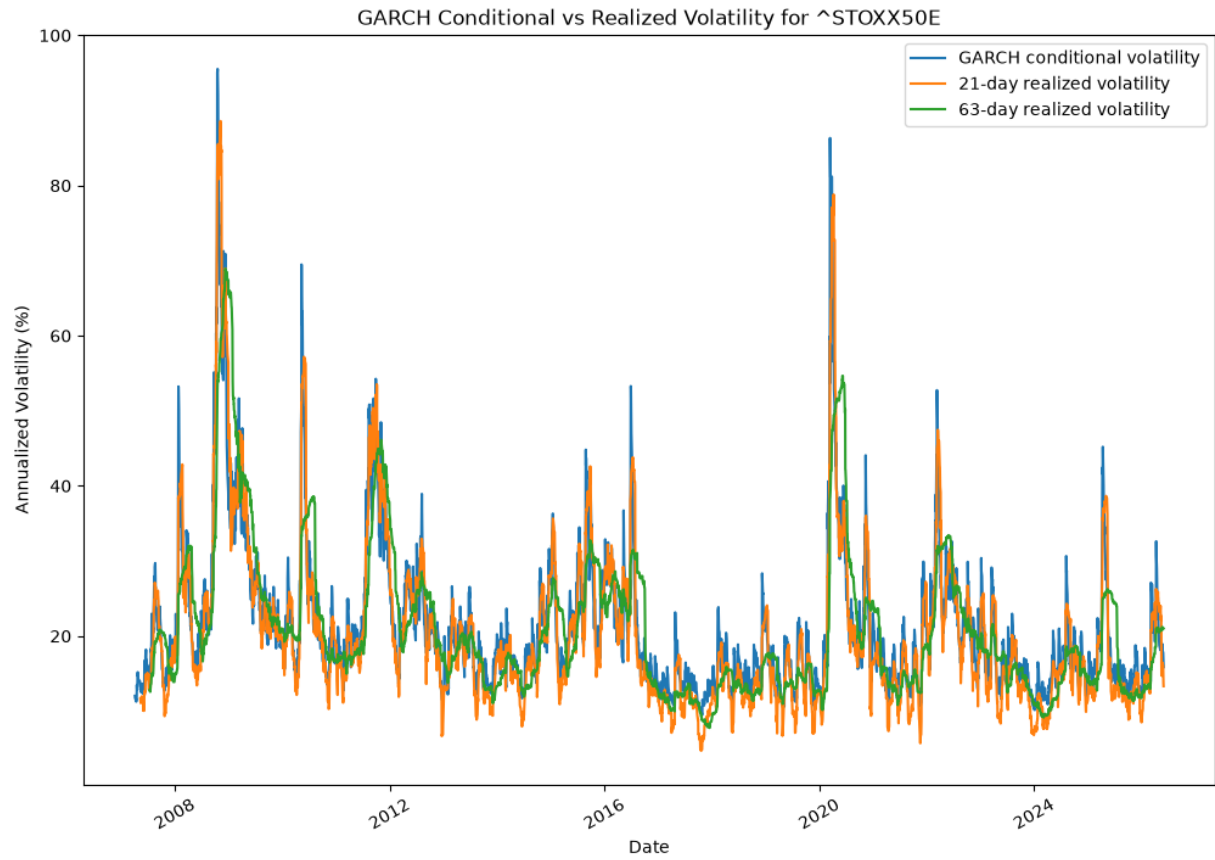
8. Volatility Forecasting

8.1 Conditional and Realized Volatility



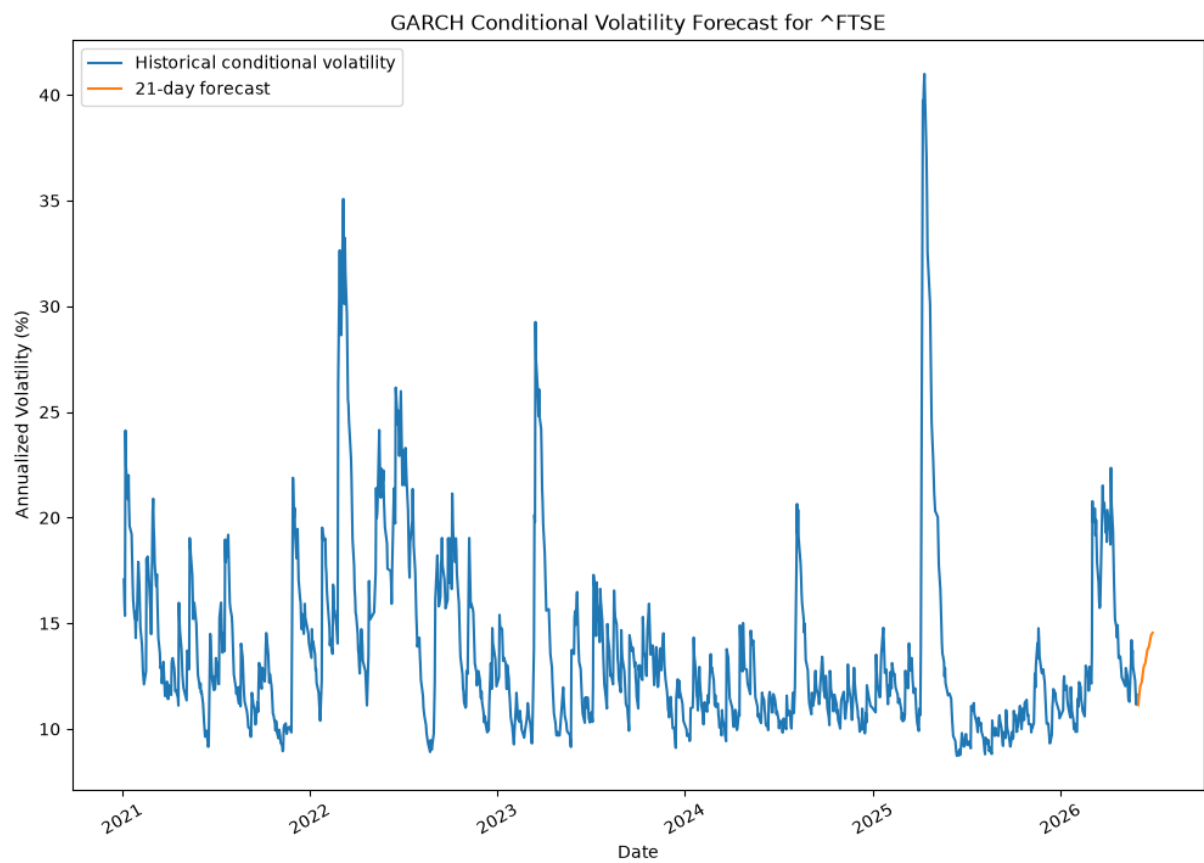


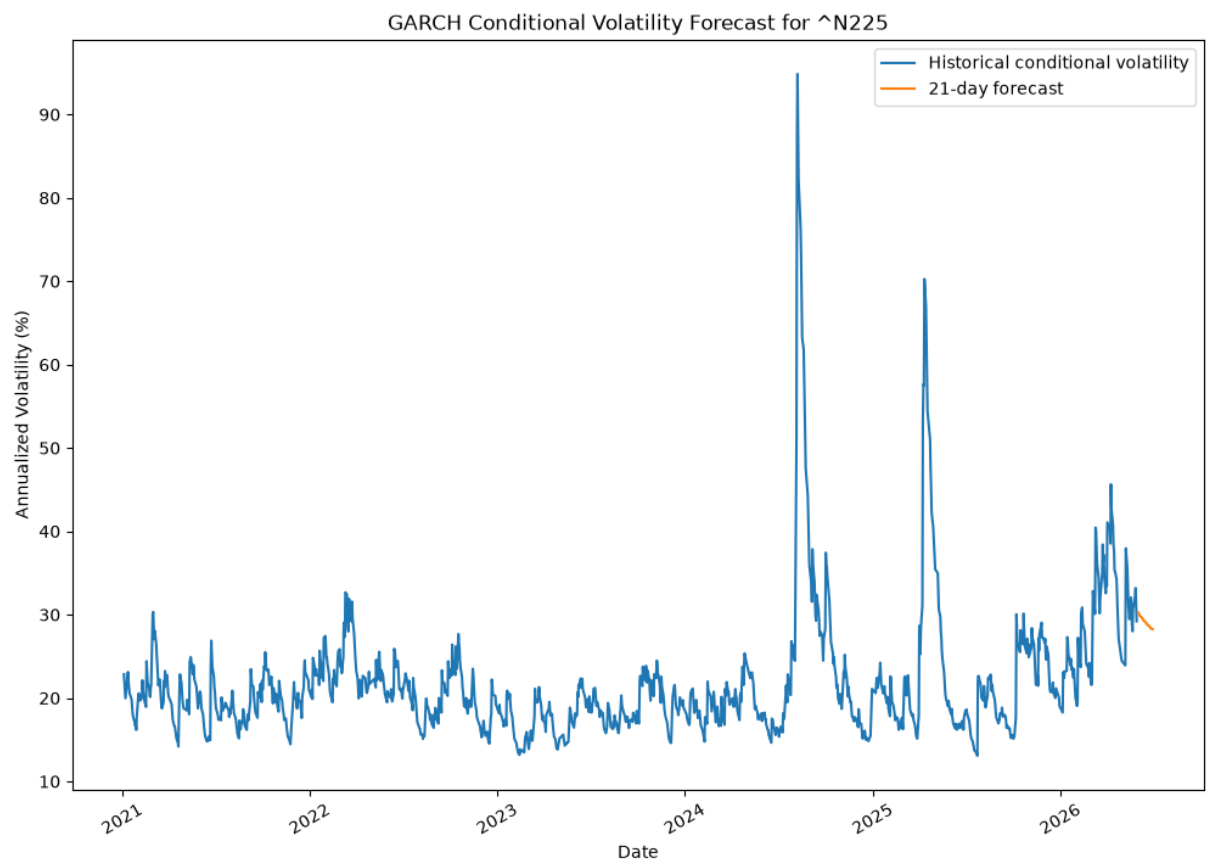


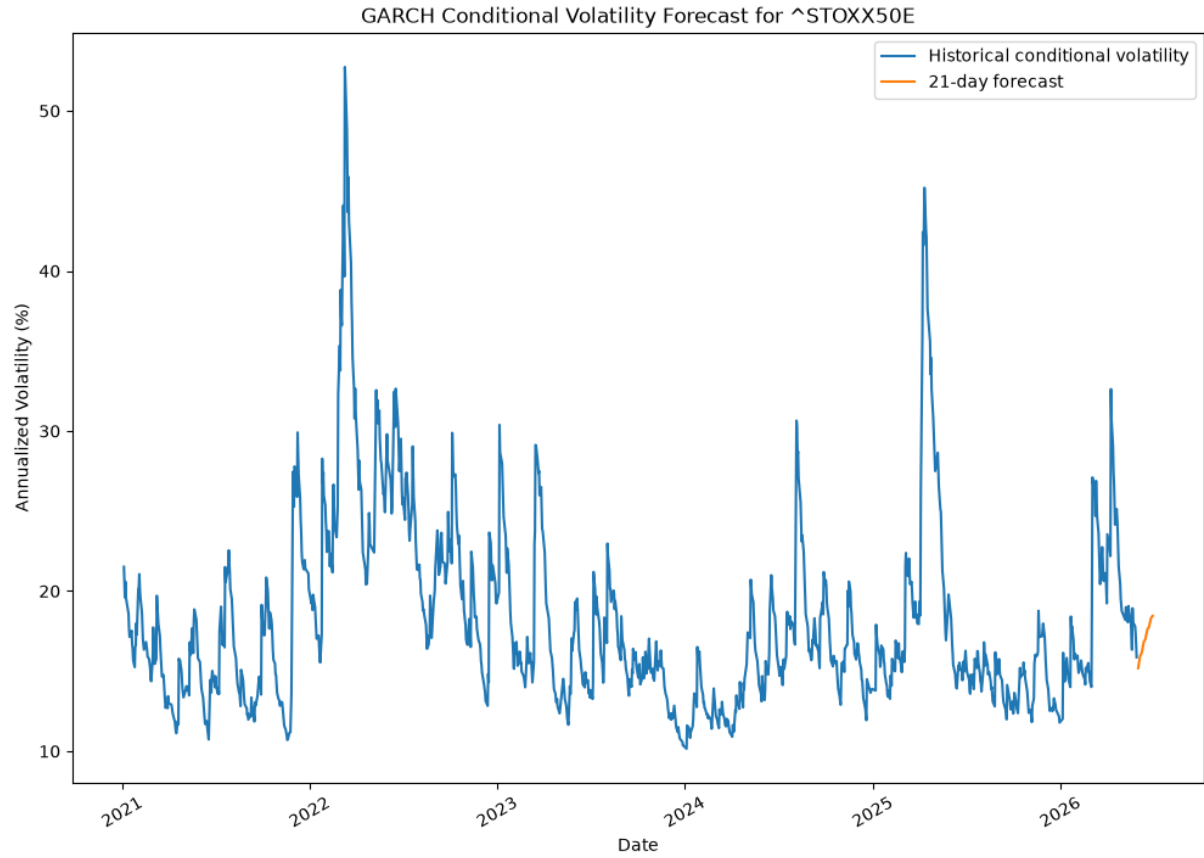


8.2 Forecasts

index	h	mean	variance	volatility	ann_volatility
$\hat{\text{FTSE}}$	21	0.0518	0.8398	0.9164	14.5478
$\hat{\text{GSPC}}$	21	0.1044	0.8293	0.9106	14.4561
$\hat{\text{N225}}$	21	0.0803	3.1675	1.7797	28.2525
$\hat{\text{STOXX50E}}$	21	0.0738	1.3499	1.1619	18.4439



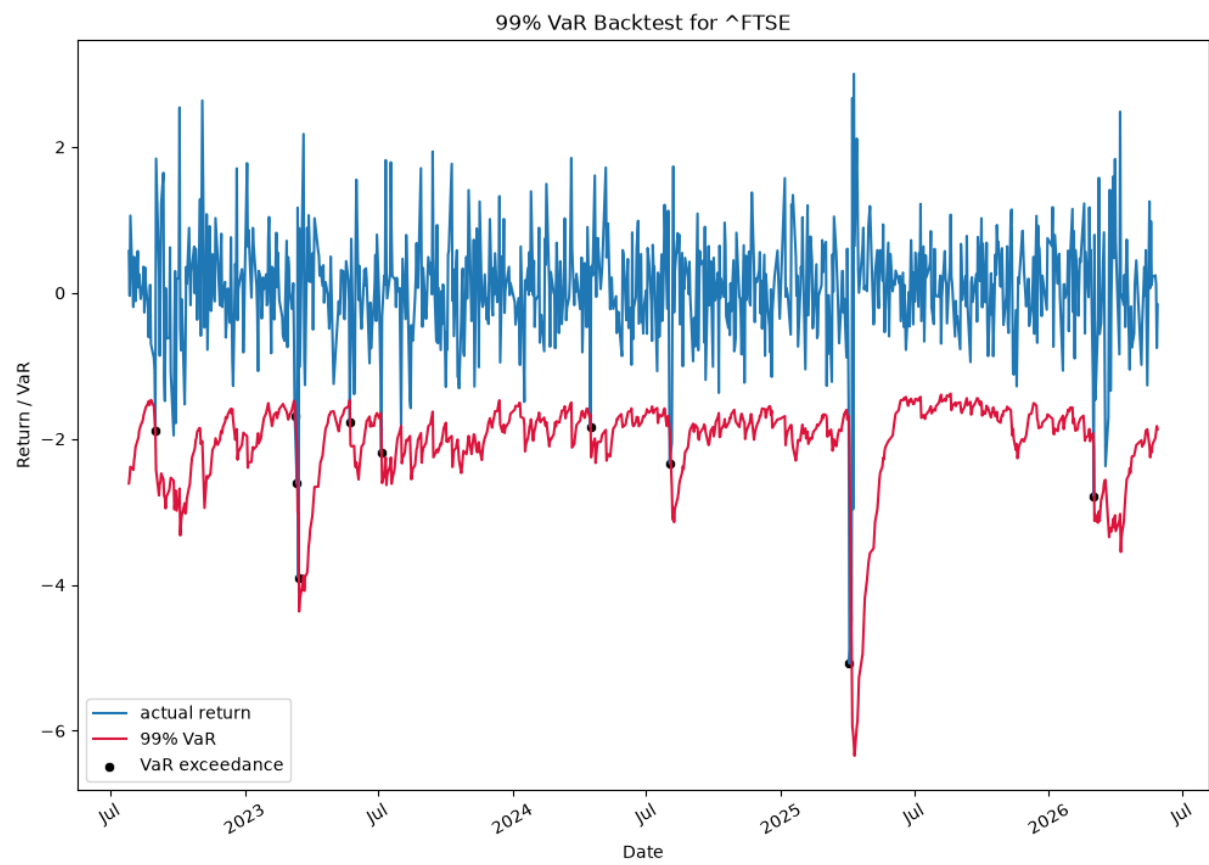


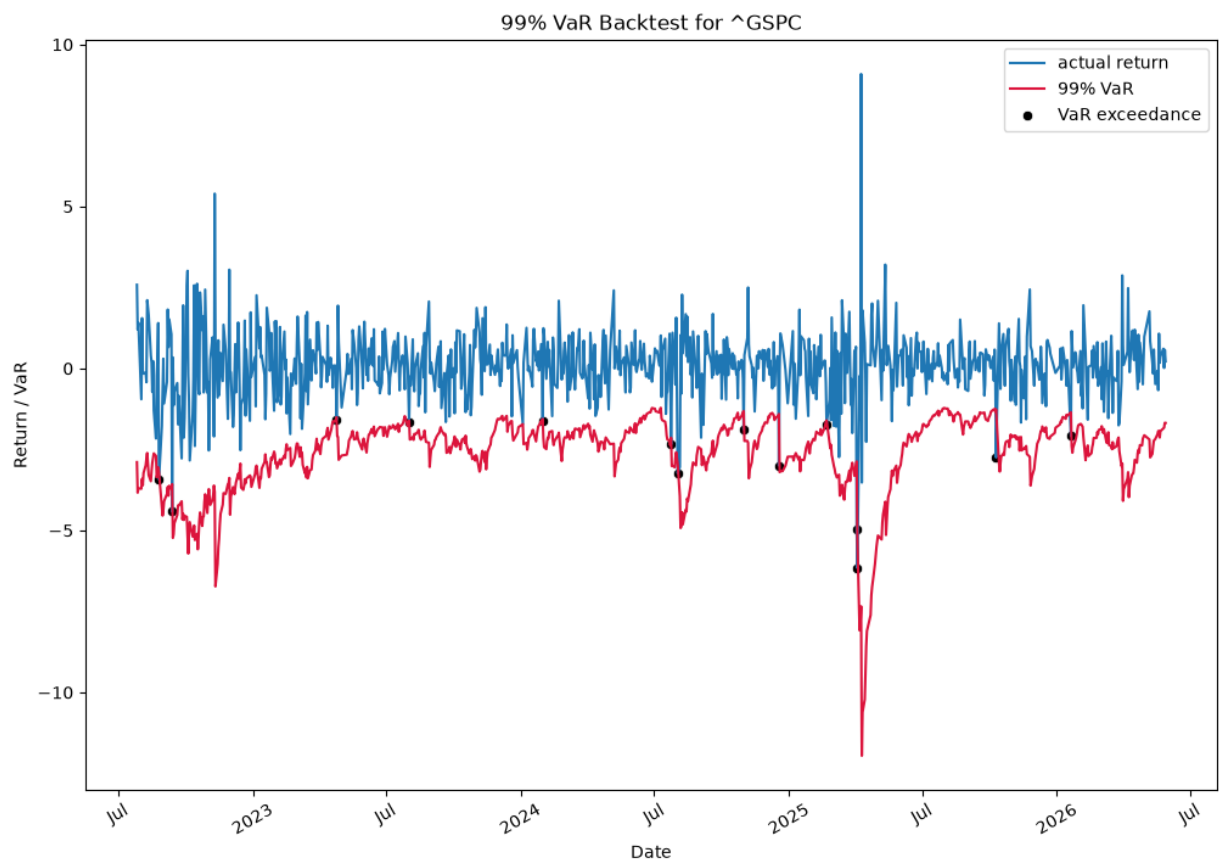


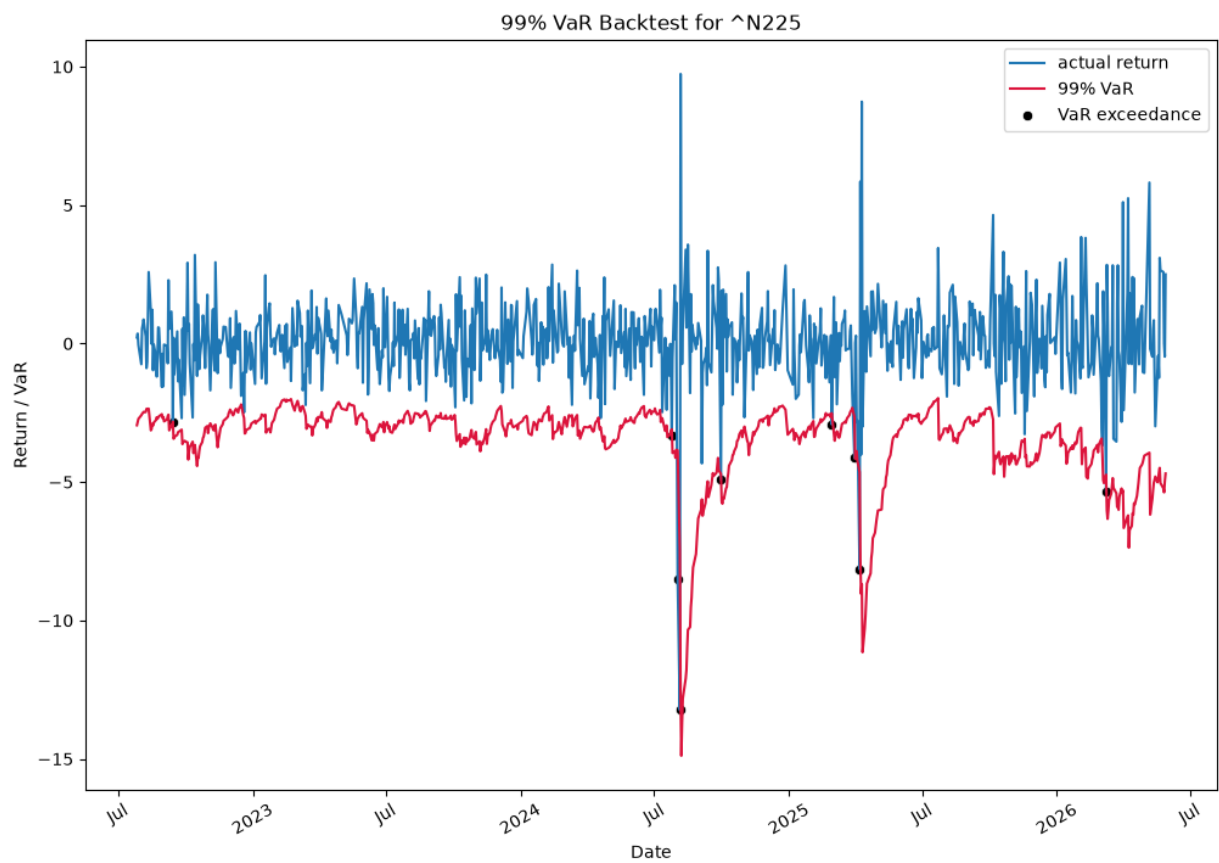
9. VaR Backtesting

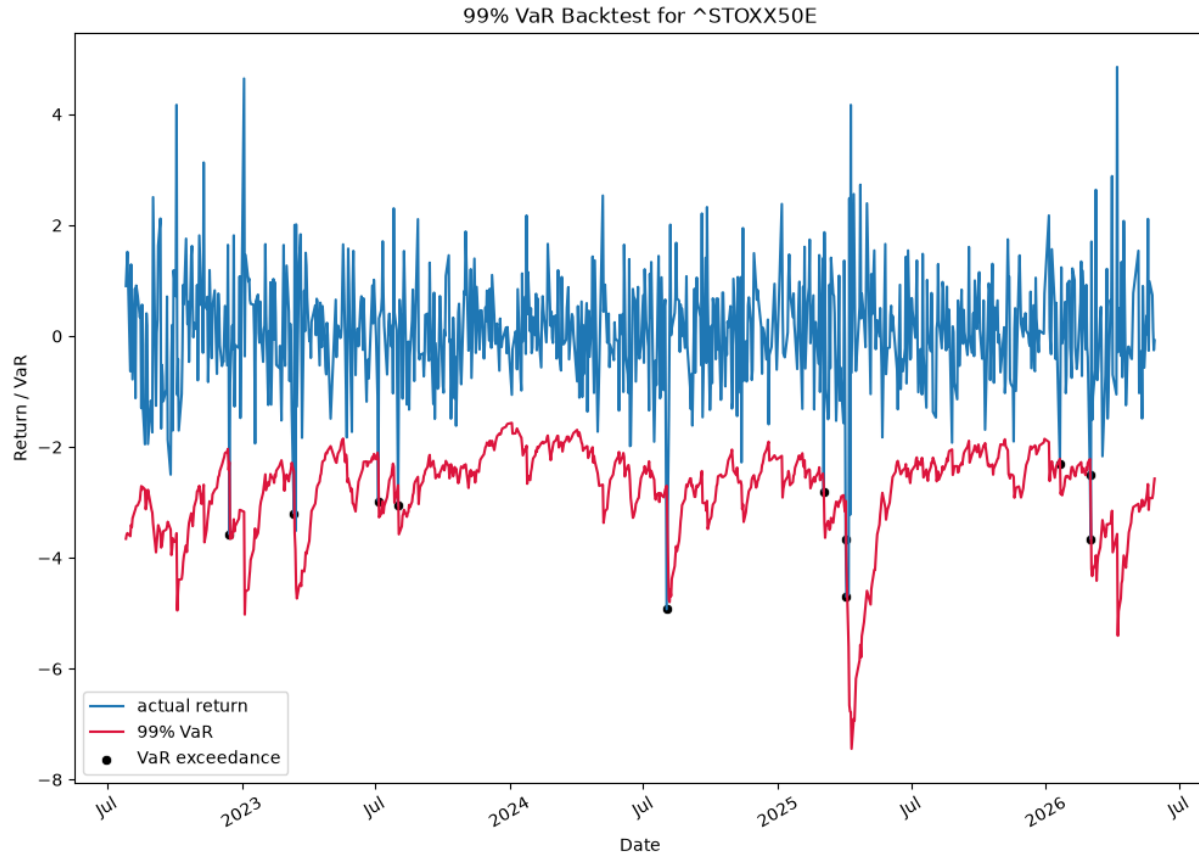
The 99% VaR backtesting using an 80%-20% train-test data split shows that the model's VaR forecasts are broadly well-calibrated for all four indices. However, ^GSPC has the most exceedances and is the closest to potential underestimation of tail risk.

index	var_level	obs	exceedances	expected	rate	kupiec_lr	kupiec_p
^FTSE	99%	875	10	8.75	0.0114	0.1724	0.678
^GSPC	99%	875	14	8.75	0.016	2.692	0.1009
^N225	99%	875	9	8.75	0.0103	0.0071	0.9326
^STOXX50E	99%	875	11	8.75	0.0126	0.5404	0.4623









10. Limitations

One limitation of the project is that the $AR(1)$ -GARCH(1, 1) model does not fully remove the autocorrelation of the squared standardized residuals for $\hat{GSPC}$ and $\hat{N225}$. To address this, future projects could compare richer volatility models such as GJR-GARCH, EGARCH, or higher-order GARCH specifications. Another limitation is that each index is modeled separately, so the analysis does not capture cross-market dependence or spillover effects across global equity markets.